# Sales Proposal Preparation System

A Multi-Agent AI Workflow built in n8n for Vertex Consulting

**Project 3: Multi-Agent AI System (Sales Department)**

WRITTEN ASSESSMENT REPORT

| | |
|---|---|
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# Table of Contents

1. Problem Understanding ..... 5

1.1 About Vertex Consulting ..... 5

1.2 The main business problem ..... 5

1.3 Clients are the input, not the problem ..... 5

1.4 The complete chain ..... 5

1.5 Why each piece of project information is needed ..... 6

2. Existing Process ..... 7

3. AI Opportunity ..... 8

4. Solution Design ..... 9

4.1 Overview ..... 9

4.2 Why the system was designed this way ..... 9

4.3 Tools used ..... 9

4.4 How the business rules are handled ..... 10

5. Multi-Agent Design ..... 12

5.1 The agents ..... 12

5.2 How the agents work together ..... 13

5.3 Why this split makes sense ..... 13

6. Workflow ..... 14

6.1 Workflow diagram ..... 14

6.2 Step-by-step flow ..... 14

7. Working System ..... 16

7.1 System demonstration video ..... 16

7.2 How to run the system ..... 16

7.3 Screenshots ..... 17

8. Testing and Improvement ..... 23

8.1 Testing approach ..... 23

8.2 Test records ..... 24

8.3 Problems found and improvements ..... 42

8.4 Test summary ..... 43

9. Final Output ..... 44

The proposal below is the actual revised output produced by the system in Test 10 (GreenPath Education), after the salesperson declined the first draft and asked for the payment terms to be stated clearly. The Revision Agent made only that one change, as shown below. ..... 44

Draft Sales Proposal: GreenPath Education ..... 44

1. Client Requirement ..... 44

2. Understanding of the Problem ..... 44

3. Recommended Solution ..... 44

4. How the Solution Will Work ..... 44

5. Scope and Deliverables ..... 44

6. Expected Benefits ..... 45

7. Estimated Timeline ..... 45

8. Estimated Cost ..... 45

9. Assumptions and Limitations ..... 45

10. Next Step ..... 45

9.1 Review result and salesperson decision ..... 46

10. Business Impact ..... 47

11. Limitations ..... 48

12. Human Oversight ..... 49

13. Conclusion ..... 50

Appendix A: Vertex Service Catalogue and Approved Prices ..... 51

Appendix B: Expected Results for the Eight Clients ..... 52

Appendix C: Agent Prompts ..... 54
C.1 Supervisor Agent ..... 54
C.2 Requirement Analyst ..... 54
C.3 Service Matcher ..... 55
C.4 Budget & Timeline Checker ..... 56
C.5 Proposal Writer ..... 57
C.6 Proposal Reviewer ..... 58
C.7 Proposal Revision Agent ..... 60
Appendix D: Testing ..... 62

# 1. Problem Understanding

## 1.1 About Vertex Consulting

Vertex Consulting is a company that helps other organizations use AI and automation in their everyday work. Its aim is to reduce repetitive work, make information easier to find, and help employees make better decisions. Vertex offers eight services, for example an AI Knowledge Assistant, an AI Customer Support Assistant and AI Reporting Automation. The full list with prices is in Appendix A.

## 1.2 The main business problem

This project has one main business problem:

Vertex salespeople spend too much time manually turning client inquiries into suitable sales proposals.

Every client inquiry is different. Each client has a different problem, budget, timeline and expectation. Before a salesperson can write a proposal, the salesperson has to:

- understand what the client really needs,
- find the Vertex service that fits,
- check the correct price,
- check that the price fits the client budget and the delivery time fits the client timeline,
- follow the company business rules, and
- write the proposal in the required format.

At the moment most of this is done by hand. This takes time and can cause mistakes, such as quoting a wrong price, promising something Vertex does not offer, or forgetting to mention a problem with the budget.

## 1.3 Clients are the input, not the problem

The school, hospital, travel company and other organizations in the project data are **clients of Vertex**. Their problems are the test inputs for the system. The real problem that this project solves belongs to the Vertex sales team: preparing a good proposal quickly and safely.

## 1.4 The complete chain

1. A client has a problem.

2. The client contacts Vertex.
3. A Vertex salesperson needs to work out what to offer.
4. The AI system helps the salesperson: it understands the client, finds the Vertex service, checks price, budget and timeline, creates the proposal and checks it.
5. The salesperson reviews the proposal and makes the final decision.
6. Only after approval can the proposal go to the client.

### 1.5 Why each piece of project information is needed

| Project information | Why the system needs it |
|---|---|
| Client requirement | To understand what the client needs |
| Client industry and company size | To give context about the client |
| Client budget | To check if the solution is affordable |
| Client timeline | To check if delivery is possible in time |
| Service catalogue | To know what Vertex can actually sell |
| Pricing | To know how much to quote |
| Business rules | To stop the AI from making bad decisions |
| Company information | To describe Vertex correctly |
| Proposal requirements | To know what sections the proposal must have |
| Multiple agents | To divide the work into clear jobs |
| Quality checker | To catch mistakes before a human sees the proposal |
| Salesperson | To make the final human decision |

# 2. Existing Process

Without the Multi-Agent System, a salesperson would normally handle each inquiry manually in the steps below.

| Step | What the salesperson does | Problem with doing it manually |
|---|---|---|
| 1 | Reads the client inquiry and notes the main facts (problem, budget, timeline, contact). | Inquiries can be short or unclear. Details can be missed or misunderstood. |
| 2 | Looks through the service catalogue to find a matching service. | Takes time, and different people may choose different services for the same inquiry. |
| 3 | Looks up the price and compares it with the client budget. | Easy to make a small mistake in prices or totals. |
| 4 | Compares the client timeline with the service delivery time. | A conflict may be noticed late or not at all. |
| 5 | Checks the business rules (no discounts, no unsupported promises, no legal commitments). | Rules are easy to forget when the salesperson is busy. |
| 6 | Writes the proposal with all the required sections. | Slow. The quality and style change from person to person. |
| 7 | Reads the proposal again and fixes mistakes, then sends it. | Mistakes can still reach the client. |

In short, the manual process is slow, repeats the same work for every inquiry, and depends a lot on the person doing it. The project data does not give exact time figures, so this report does not invent any.

# 3. AI Opportunity

Not every step needs AI. AI is best where the work needs reading, understanding and writing. Normal workflow automation is better where the work is exact, such as looking up a value or doing a calculation, because it gives the same correct result every time.

| Task | Best approach | Why |
|---|---|---|
| Read the inquiry and pull out the key facts | AI | Inquiries are written in normal language and can be unclear. |
| Spot missing or unclear information | AI | Needs understanding of what is stated and what is not. |
| Match the client problem to a service | AI (using the catalogue) | Needs judgement, but only from the approved catalogue. |
| Find the client record and service price | Workflow automation | It is a simple lookup. No guessing is allowed. |
| Compare price with budget and delivery time with timeline | Workflow automation (exact check) | Numbers must be exact. The AI then explains the result in words. |
| Write the proposal in clear business language | AI | Writing is what AI does well. |
| Check the proposal against the rules | AI reviewer with a fixed checklist | Catches invented prices, promises and missing sections. |
| Send the review email and wait for the answer | Workflow automation | It is a fixed process that needs no AI. |
| Create the Google Doc and send the final email | Workflow automation | It is the same every time. |
| Approve or decline the proposal | Human (salesperson) | The rules say a salesperson must decide. |

# 4. Solution Design

## 4.1 Overview

The solution is an n8n workflow that starts from a client inquiry and ends with a proposal waiting for salesperson approval. Several AI agents each do one clear job. The output of one agent becomes the input of the next. Before the proposal reaches a human, a Proposal Reviewer agent checks it. The salesperson can then approve it or decline it with comments. If declined, a Revision Agent changes the proposal and the review is repeated.

## 4.2 Why the system was designed this way

**One job per agent.** A single big agent that does everything would mix up tasks and would be hard to test. With separate agents, each step can be checked on its own.

**Stay inside the provided data.** Agents only receive the client record, catalogue, prices, rules and proposal requirements. They are told not to invent services, prices or client needs.

**Exact work is not left to AI.** Prices, totals, budget and timeline comparisons are exact checks.

**Check before the human.** The Proposal Reviewer catches mistakes so the salesperson sees a cleaner draft.

**Human decision is compulsory.** The workflow stops and waits for the salesperson. Nothing is sent to the client without approval.

**A repair loop.** A declined proposal goes back to be fixed instead of the whole process starting again.

## 4.3 Tools used

| Tool | Use in the project |
|---|---|
| n8n | Builds and runs the whole workflow. |
| Chat trigger (When chat message received) | Starts the workflow when a client inquiry is entered. |
| OpenRouter chat models | The language models for the agents. Model used:openai/gpt-4.1-mini |
| Supervisor Memory | Keeps the short conversation history for the Supervisor Agent. |

| Tool | Use in the project |
|---|---|
| Gmail (Send and Wait for response) | Sends the proposal to the salesperson and waits for Approve or Decline plus comments. |
| Google Docs | Creates the final proposal document after approval. |
| Data source | The client records, catalogue, prices and rules are stored in prompt text |

### 4.4 How the business rules are handled

| Business rule | How the system follows it |
|---|---|
| Use only approved prices | Prices come only from the approved price list. Agents are told never to create a price. |
| AI cannot offer or approve discounts | If a discount is requested, the system says it cannot offer one and flags it for the salesperson. |
| Significant customization needs salesperson review | The Budget & Timeline Checker and Reviewer flag any customization. |
| Stay within client budget | The budget check reports any gap. Options are suggested, but never a discount. |
| Delivery time must match service information | Only the catalogue delivery time is used. Conflicts are flagged. |
| Do not promise services Vertex does not offer | The Service Matcher can choose only from the catalogue. The Reviewer checks this again. |
| No legal or contract commitments | The proposal is marked as a draft with no binding commitment. |

| Business rule | How the system follows it |
| --- | --- |
| Standard payment terms (50% at start, 50% after completion) | The Writer states these standard terms in the cost section. |
| Salesperson must approve before sending | Gmail Send and Wait pauses the workflow until the salesperson decides. |
| Use client information only for the relevant proposal | Each run handles one client a separate memory session |

# 5. Multi-Agent Design

## 5.1 The agents

The system has seven agents. Each has a different responsibility, so they do not repeat each other's work.

| Agent | Responsibility | Receives | Produces |
|---|---|---|---|
| Supervisor Agent | Coordinates the work. Keeps track of the current stage and passes the inquiry on. It does not analyse, write or decide content. | The client inquiry from the chat | A short status and the inquiry for the next agent |
| Requirement Analyst | Turns the inquiry into clear facts: problem, goal, budget, timeline, contact. Lists anything missing or unclear. Does not assume anything. | Client record and inquiry | Structured client analysis with a missing-information list |
| Service Matcher | Chooses the best service or services from the catalogue and gives the reasons. | Client analysis and service catalogue | Recommended service with reasoning |
| Budget & Timeline Checker | Compares approved price with budget and service delivery time with client timeline. Checks business rules and lists conflicts and flags. | Recommended service, approved prices, rules | Budget and timeline result, conflicts, salesperson flags |
| Proposal Writer | Writes the proposal with the 10 required sections in simple business language. | All earlier outputs and proposal requirements | Draft proposal |

| Agent | Responsibility | Receives | Produces |
|---|---|---|---|
| Proposal Reviewer | Checks the draft for invented prices, unsupported promises, guarantees, missing sections, wrong deliverables and unflagged assumptions. | Draft proposal, catalogue, rules | Review result and a list of issues |
| Proposal Revision Agent | Fixes the proposal using the salesperson comments or reviewer issues, changing only what is needed. | Draft proposal and comments | Revised proposal |

### 5.2 How the agents work together

The agents work like a small team. The Supervisor receives the inquiry and starts the chain. The Requirement Analyst understands the client. The Service Matcher picks the service. The Budget & Timeline Checker tests it against the money, time and rules. The Proposal Writer turns everything into a proposal. The Proposal Reviewer checks it. The salesperson then reviews the reviewed proposal.

If the salesperson declines, the comments go to the Proposal Revision Agent, which fixes the proposal and sends it back to the Reviewer and the salesperson. This continues until the proposal is approved.

### 5.3 Why this split makes sense

Sales work already has natural stages: understand, match, check, write, review. Giving each stage to its own agent means each agent has a small, clear task and a limited input. This reduces mistakes, and makes it easy to see which stage caused a problem when testing. The Analyst finds facts, the Matcher makes a choice, the Checker tests limits, the Writer writes, and the Reviewer criticizes. These are different skills, not the same task repeated.

# 6. Workflow

## 6.1 Workflow diagram

Figure 1 shows the workflow in n8n.

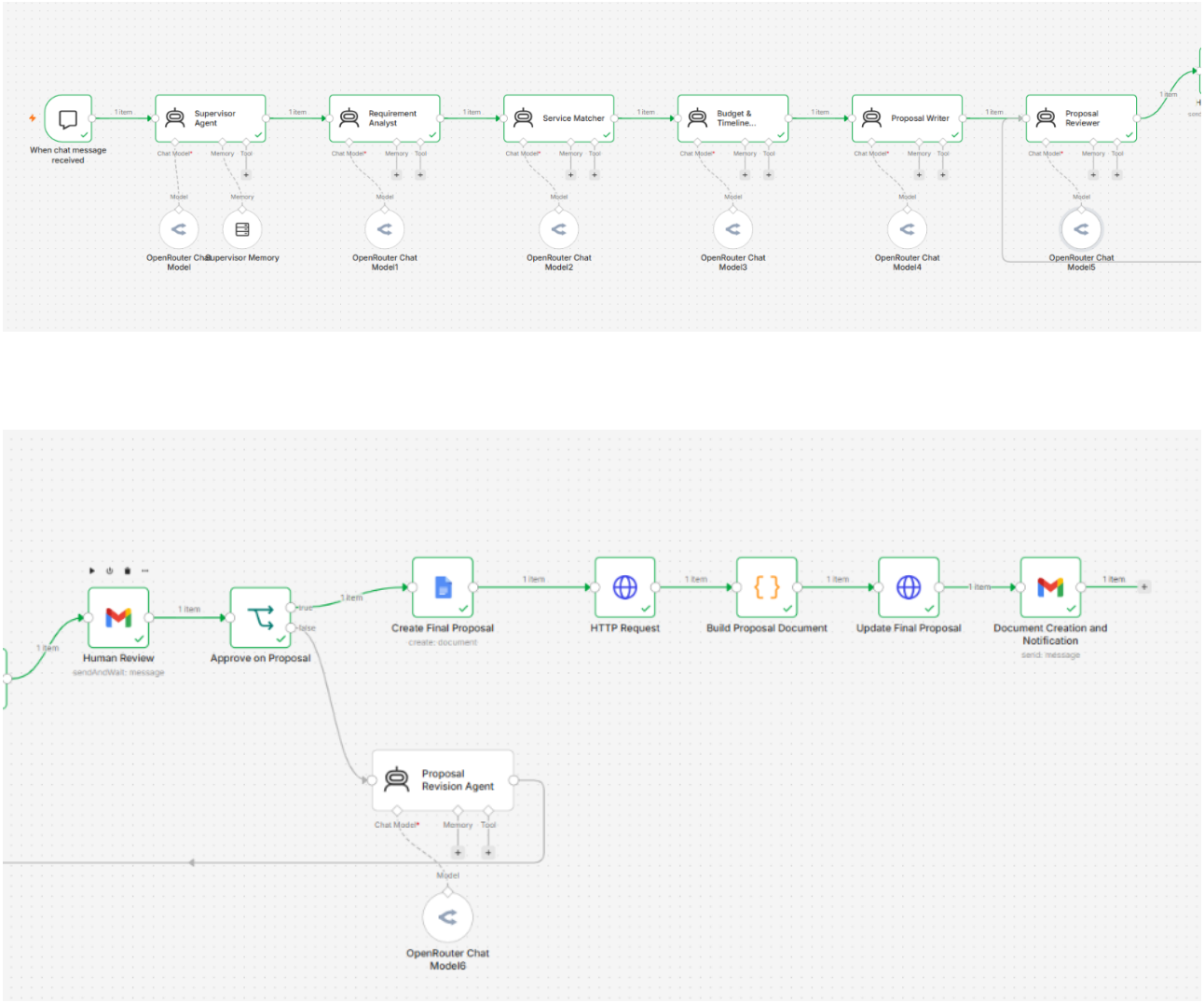



*Figure 1: n8n workflow of the Sales Proposal Preparation System*

## 6.2 Step-by-step flow

| Step | Node | What happens |
|---|---|---|
| 1 | When chat message received | A client inquiry is entered and the workflow starts. |

| Step | Node | What happens |
|---|---|---|
| 2 | Supervisor Agent | Receives the inquiry, starts the process and tracks the stage. |
| 3 | Requirement Analyst | Extracts the client facts and lists missing or unclear information. |
| 4 | Service Matcher | Selects the matching Vertex service(s) from the catalogue. |
| 5 | Budget & Timeline Checker | Checks price against budget, delivery against timeline, and business rules. |
| 6 | Proposal Writer | Writes the draft proposal in the 10 required sections. |
| 7 | Proposal Reviewer | Quality check of the draft against the rules and catalogue. |
| 8 | Human Review (Gmail Send and Wait) | Emails the proposal to the salesperson with an Approve/Decline form and a comment box. The workflow waits. |
| 9 | Approve on Proposal (If) | Reads the salesperson's decision. |
| 10a | If Approved | The system creates the final proposal as a Google Doc, then sends a confirmation email to the salesperson with a link to the finished document. |
| 10b | If Declined | The Proposal Revision Agent updates the proposal using the salesperson's comments. It goes back to the Proposal Reviewer and the salesperson again. |

The proposal is never sent directly to the client. The approved document is given to the salesperson, who sends it.

# 7. Working System

## 7.1 System demonstration video

A recording of the working system is available at the link below.

**Video link:**

https://drive.google.com/file/d/1txSlU6AxQRY-nr7aqrcbJtiOGXNPE8vh/view?usp=drive_link

The video shows:

the original client inquiry being entered,

each agent producing its output,

the review email received by the salesperson,

a decline with comments and the revised proposal,

the approved proposal in Google Docs, and

tests where the client situation changes (budget conflict, timeline conflict, missing information).

## 7.2 How to run the system

1. Open the workflow in n8n and make sure the credentials (OpenRouter, Gmail, Google Docs) are connected.
2. Open the chat window and enter a client inquiry. Example input:

   Client: Sunrise Hospital Industry Healthcare

   Company Size 200 employees

   Client Requirement Our administration team wants a better way to organize internal documents and quickly find information from policies and procedures.

   Budget NPR 250,000

   Preferred Timeline 2 months

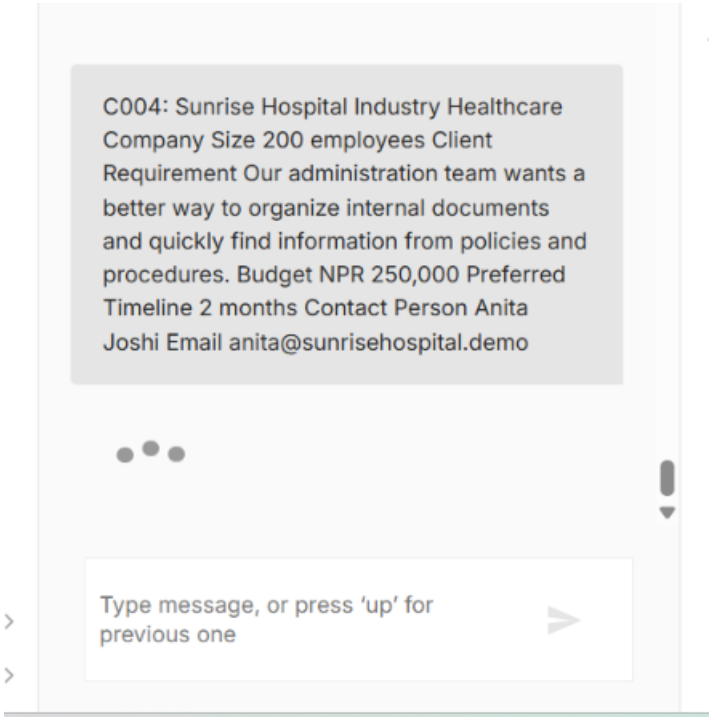
   Contact Person Anita Joshi

   Email anita@sunrisehospital.demo
3. Watch the agents run one after another in the execution view.
4. Open the email sent to the salesperson and read the proposal.

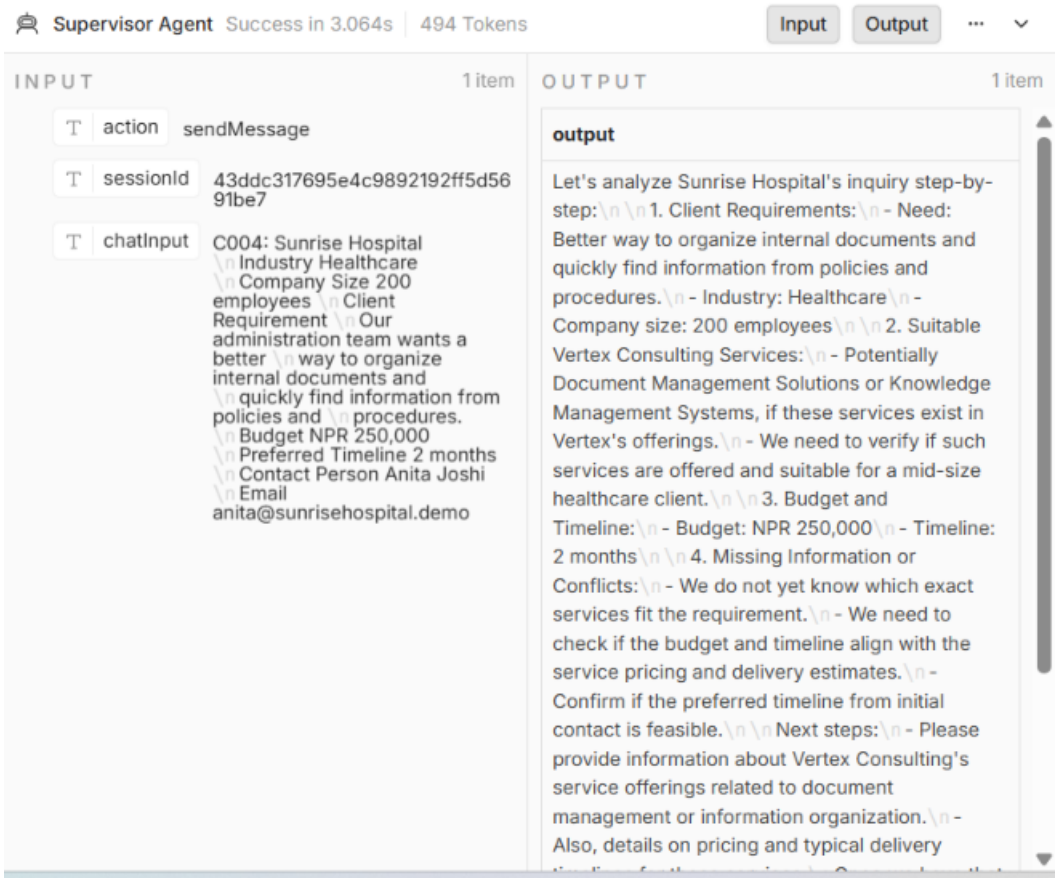
5. Click Respond and choose Approve or Decline, and add comments if needed.
6. After approval, the system creates the final proposal as a Google Doc.
7. A confirmation email is automatically sent to the salesperson with a link to the final Google Doc.

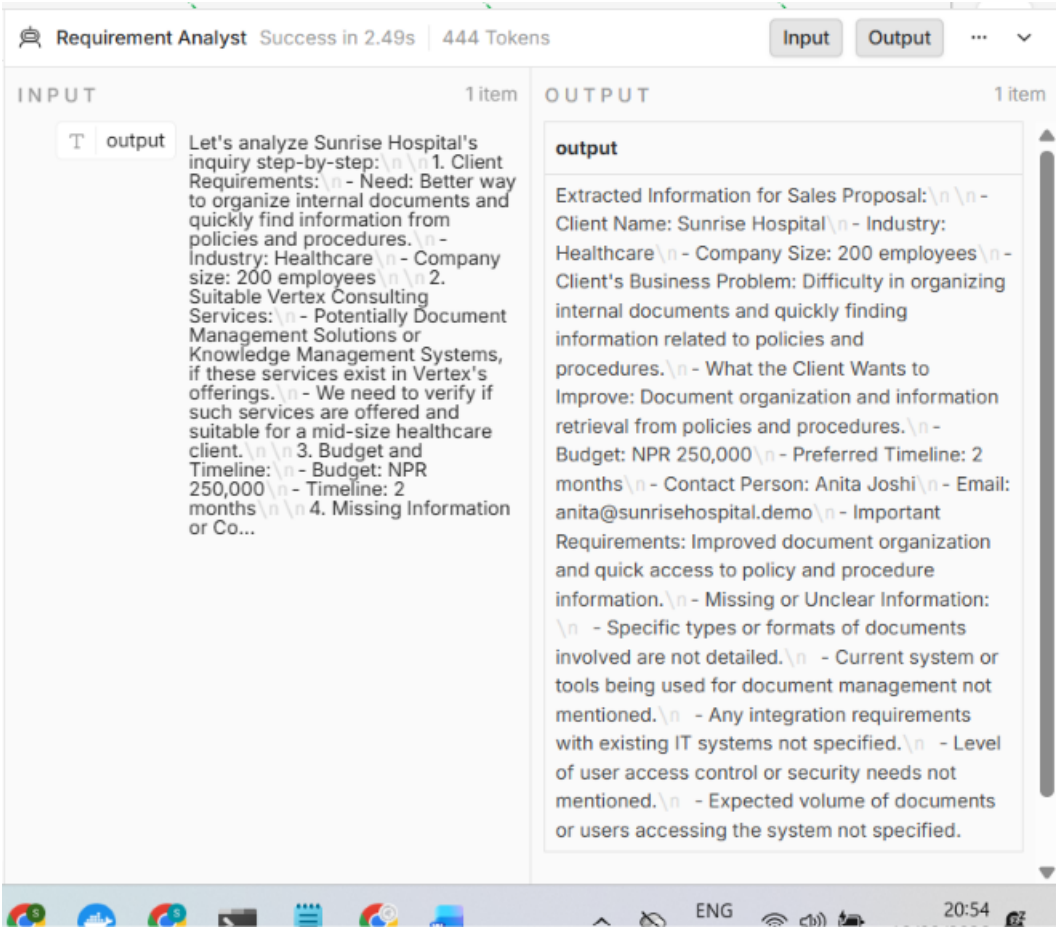
### 7.3 Screenshots

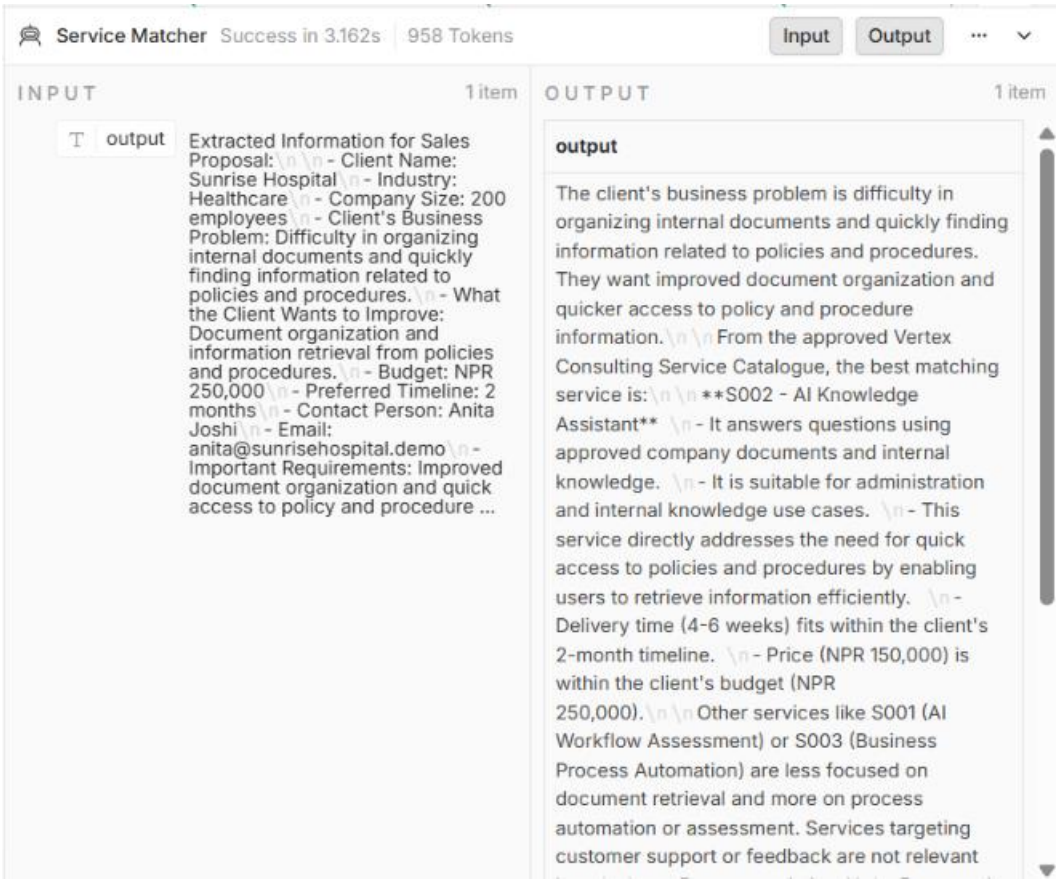
(1) chat input,

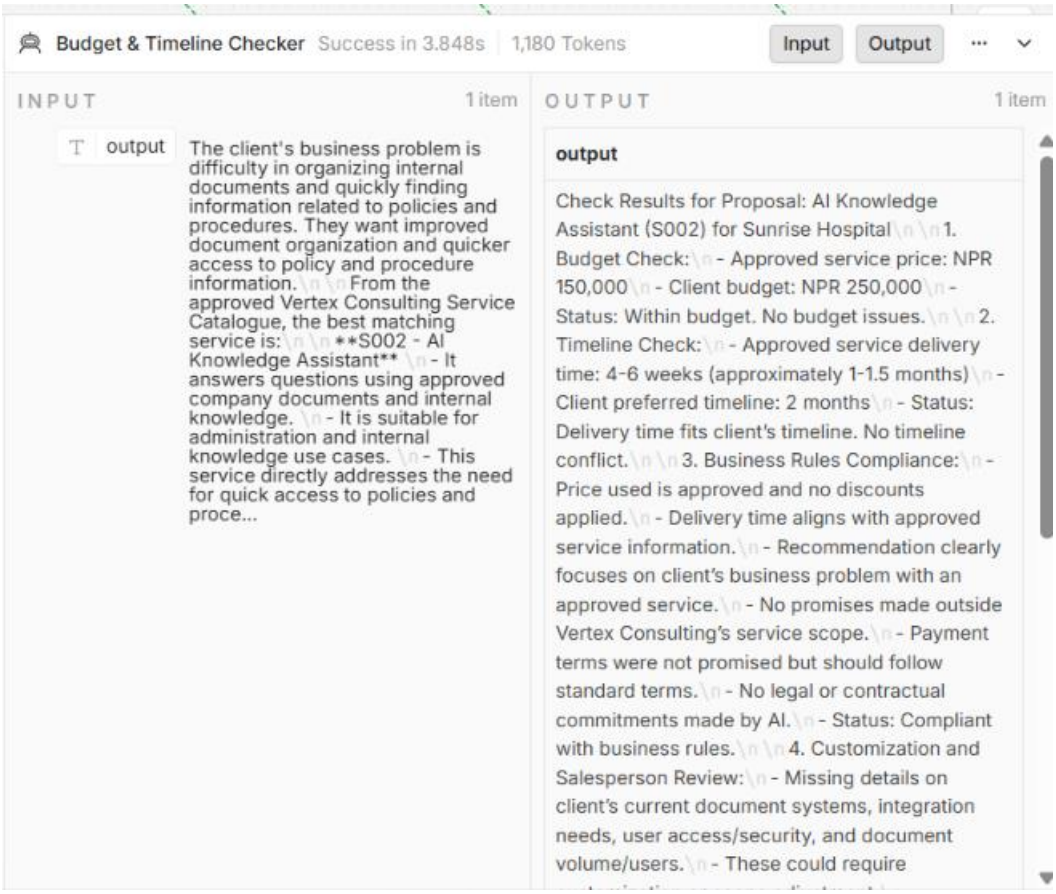


(2) an agent output such as the Supervisor Agent, Service Matcher,

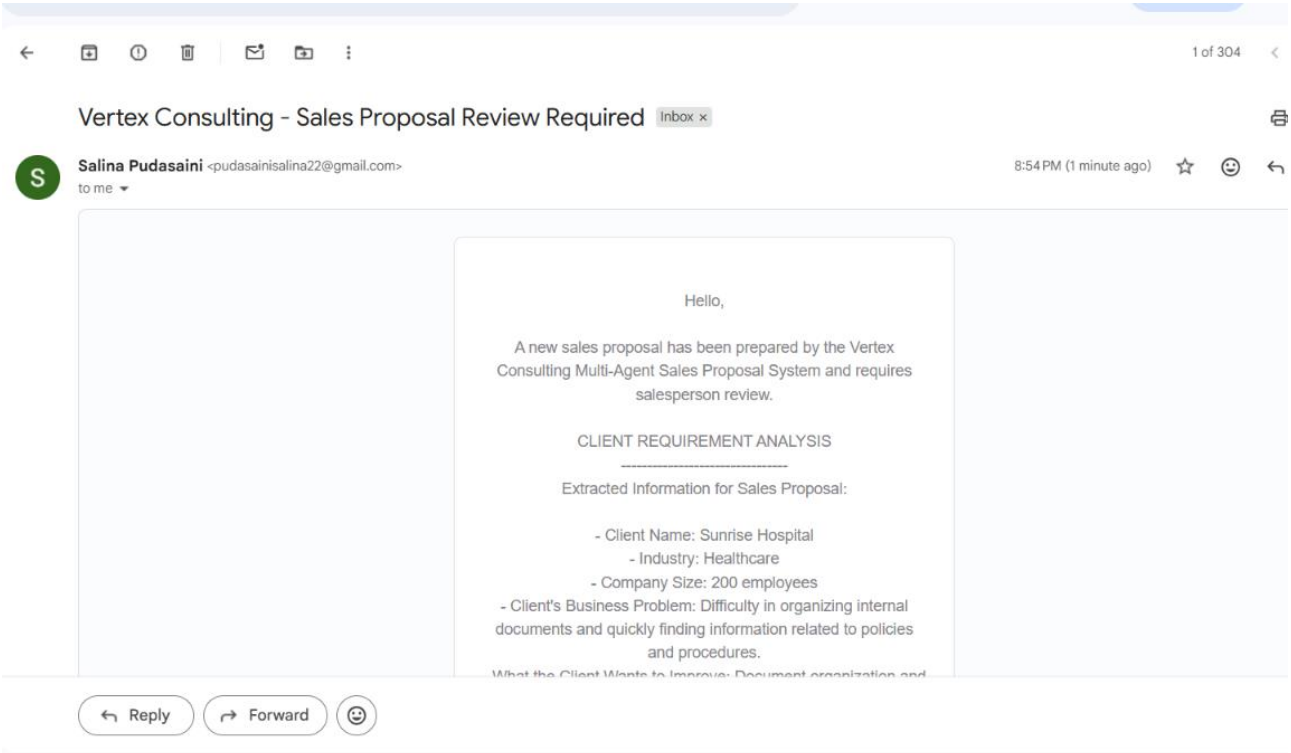



Requirement Analyst Success in 2.49s 444 Tokens
Input
Output
INPUT
1 item
output
Let's analyze Sunrise Hospital's inquiry step-by-step:\n\n1. Client Requirements:\n - Need: Better way to organize internal documents and quickly find information from policies and procedures.\n - Industry: Healthcare\n - Company size: 200 employees\n\n2. Suitable Vertex Consulting Services:\n - Potentially Document Management Solutions or Knowledge Management Systems, if these services exist in Vertex's offerings.\n - We need to verify if such services are offered and suitable for a mid-size healthcare client.\n\n3. Budget and Timeline:\n - Budget: NPR 250,000\n - Timeline: 2 months\n\n4. Missing Information or Co...
OUTPUT
1 item
output
Extracted Information for Sales Proposal:\n\n - Client Name: Sunrise Hospital\n - Industry: Healthcare\n - Company Size: 200 employees\n - Client's Business Problem: Difficulty in organizing internal documents and quickly finding information related to policies and procedures.\n - What the Client Wants to Improve: Document organization and information retrieval from policies and procedures.\n - Budget: NPR 250,000\n - Preferred Timeline: 2 months\n - Contact Person: Anita Joshi\n - Email: anita@sunrisehospital.demo\n - Important Requirements: Improved document organization and quick access to policy and procedure information.\n - Missing or Unclear Information:\n - Specific types or formats of documents involved are not detailed.\n - Current system or tools being used for document management not mentioned.\n - Any integration requirements with existing IT systems not specified.\n - Level of user access control or security needs not mentioned.\n - Expected volume of documents or users accessing the system not specified.
ENG
20:54

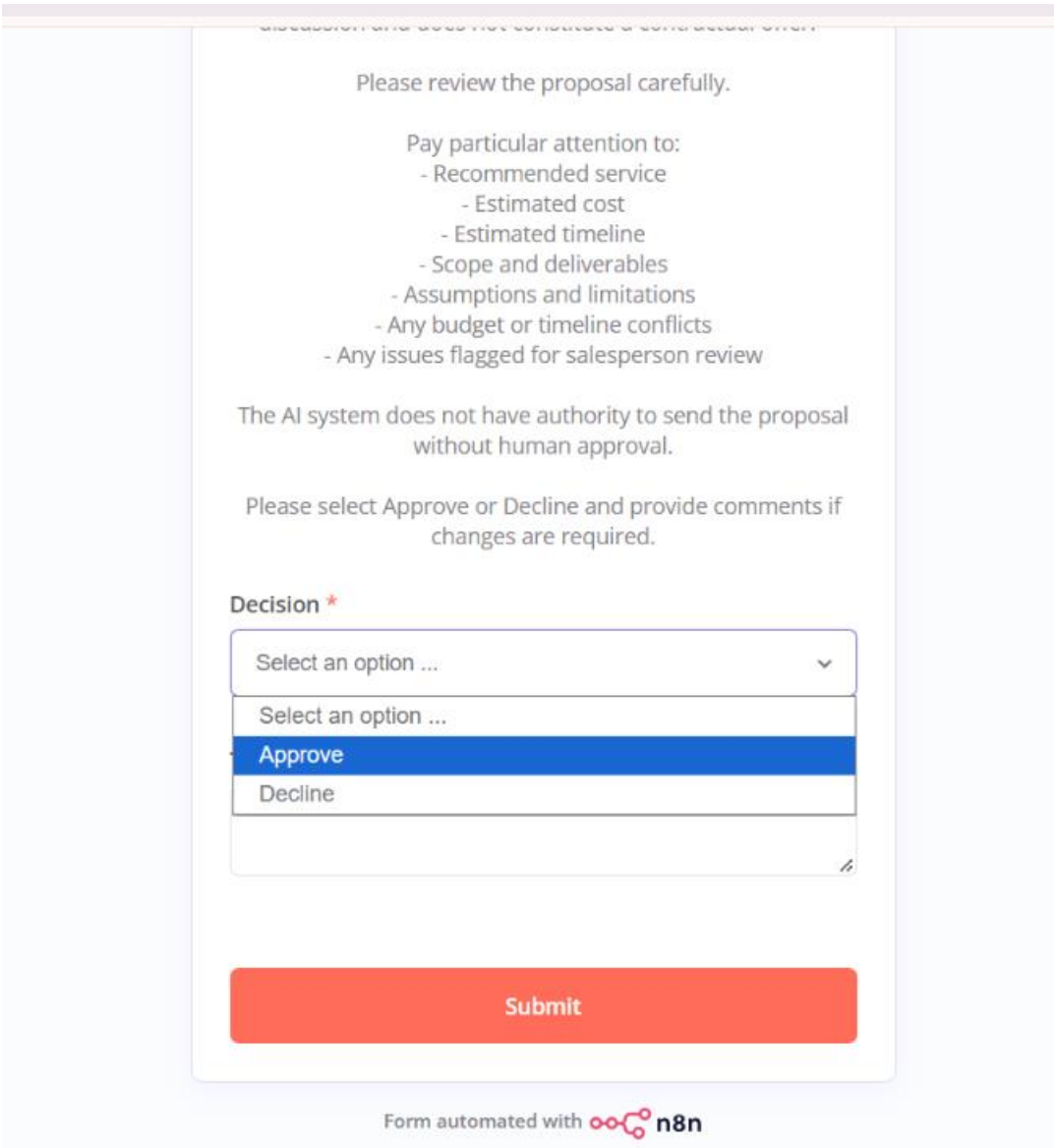

Service Matcher Success in 3.162s 958 Tokens
Input
Output
INPUT
1 item
output
Extracted Information for Sales Proposal:\n\n - Client Name: Sunrise Hospital\n - Industry: Healthcare\n - Company Size: 200 employees\n - Client's Business Problem: Difficulty in organizing internal documents and quickly finding information related to policies and procedures.\n - What the Client Wants to Improve: Document organization and information retrieval from policies and procedures.\n - Budget: NPR 250,000\n - Preferred Timeline: 2 months\n - Contact Person: Anita Joshi\n - Email: anita@sunrisehospital.demo\n - Important Requirements: Improved document organization and quick access to policy and procedure ...
OUTPUT
1 item
output
The client's business problem is difficulty in organizing internal documents and quickly finding information related to policies and procedures. They want improved document organization and quicker access to policy and procedure information.\n\nFrom the approved Vertex Consulting Service Catalogue, the best matching service is:\n\n**S002 - AI Knowledge Assistant**\n - It answers questions using approved company documents and internal knowledge.\n - It is suitable for administration and internal knowledge use cases.\n - This service directly addresses the need for quick access to policies and procedures by enabling users to retrieve information efficiently.\n - Delivery time (4-6 weeks) fits within the client's 2-month timeline.\n - Price (NPR 150,000) is within the client's budget (NPR 250,000).\n\nOther services like S001 (AI Workflow Assessment) or S003 (Business Process Automation) are less focused on document retrieval and more on process automation or assessment. Services targeting customer support or feedback are not relevant



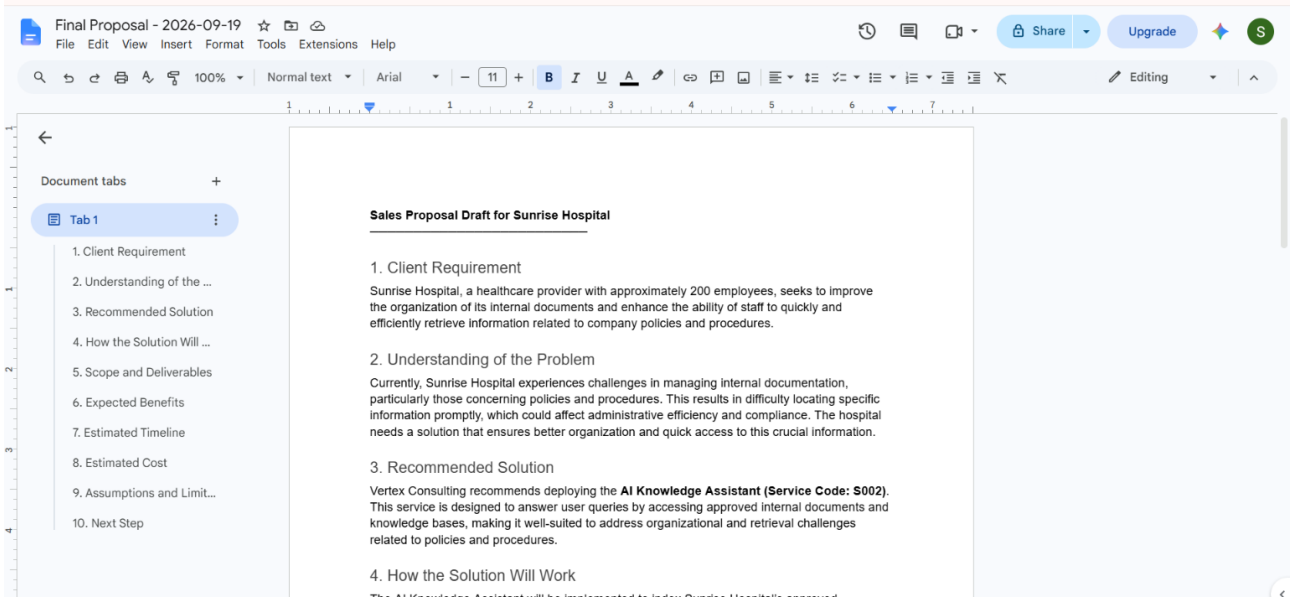

(3) the review email,



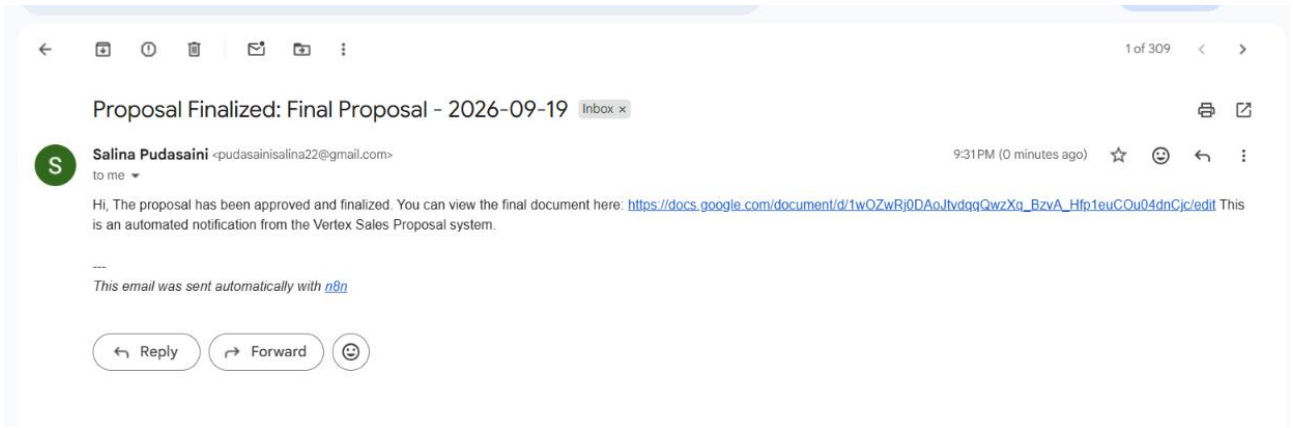

(4) the Approve/Decline form




(5) the final Google Doc




(6) the final mail



# 8. Testing and Improvement

## 8.1 Testing approach

The system was tested with more than one client and with changed situations, not just a normal case. Each test changes one thing so the result can be compared. The tests cover: a normal case, different client requirements, a budget lower than the price, a request outside the catalogue, a timeline conflict, missing information, a discount request, an unclear requirement, the decline and revision loop, and consistency of results when the same client is run again.

For each test the expected result was written before running the system. Then the actual result was compared with it.

### 8.2 Test records

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| T01 | C002 GreenPath Education. Normal case. | S004 recommended. Price NPR 175,000 within budget NPR 180,000. Delivery 4 to 6 weeks fits 6 weeks, at the top of the range with no buffer. Only catalogue deliverables. Standard payment terms shown. | Rerun recommended S002 – AI Knowledge Assistant (NPR 150,000, 4–6 weeks) instead of the originally-tested S004 – AI Customer Support Assistant (NPR 175,000). Both are within budget/timeline. Contact email now correctly captured. Reviewer passed the proposal; Human Review node recorded Decision: Approve. | (1) Service recommendation is inconsistent across identical runs same inquiry produced S004 in the first run and S002 in this rerun. (2) Deliverables still don't mirror catalogue wording exactly (training materials vs. user guide; extra deployment line). (3) Timeline section still doesn't explicitly flag "top of range, no buffer," despite this being called out as a required fix. (4) Payment terms phrased as an example ("e.g.") rather than a definite statement. | Contact-field mapping fix confirmed working. Deliverable-matching and timeline-wording instructions need to be strengthened further in the Writer Agent's system prompt — current version only partially applies them. Consider forcing the Recommender Agent to be more deterministic (e.g., lower temperature, or a stricter matching rule between problem keywords and catalogue "Suitable For" field) to fix the S002/S004 inconsistency. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| T02 | C003 Everest Traders. Reporting automation. | S005 AI Reporting Automation, NPR 160,000, within NPR 200,000. Delivery 3 to 5 weeks against 1 month flagged as a possible timeline conflict. Asks about spreadsheets and data sources. | System correctly picked S005 – AI Reporting Automation, NPR 160,000 (within the NPR 200,000 budget). It noticed the 3–5 week delivery window could slightly overshoot the client's 1-month ask and flagged this as a soft risk, not a hard conflict. Reviewer approved it with follow-up notes. | Small wording mix-up: the client's timeline is "1 month," but the proposal and reviewer both call this "4 weeks." A month is usually a bit more than 4 weeks, so this isn't wrong, but it makes the "5 weeks might overshoot" warning slightly confusing — is it overshooting a 4-week target or a ~4.3-week month? No missing info was invented, and no rules were broken. | Update the Writer/Reviewer prompts to always use the client's original wording ("1 month") instead of converting it to "4 weeks" on its own, so the timeline comparison stays clear and consistent. |

| | | | | | |
|---|---|---|---|---|---|
| T03 | C002 with budget changed to NPR 150,000. Budget lower than price. | S004 price NPR 175,000 is NPR 25,000 over budget. The issue is stated clearly. Options from the catalogue may be suggested. No discount offered. Flagged for the salesperson. | This time the system recommended S002 – AI Knowledge Assistant (NPR 150,000) instead of S004, so the cost exactly matched the new NPR 150,000 budget. No budget conflict appears anywhere in the proposal or reviewer notes. | This is the same inconsistency we saw in Test 1 — the system is switching between S002 and S004 for the same GreenPath inquiry. Here it's a problem: the test was specifically designed to check that the AI notices and flags a budget shortfall (since S004 at NPR 175,000 doesn't fit a NPR 150,000 budget). Instead, the system quietly swapped to a cheaper service that fits, without ever saying "S004 would have been over budget, so we chose an alternative." That's not wrong exactly, but it skips the whole point of the test — showing the budget conflict clearly. If a real client's | Update the Recommender Agent's instructions so that when the "obvious" service choice would exceed budget, it must explicitly say so — e.g. "S004 (NPR 175,000) exceeds the client's NPR 150,000 budget by NPR 25,000; recommending S002 instead as a lower-cost alternative." This makes the budget-check reasoning visible instead of hidden, which is what the business rules require. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | | situation only fit S004, this same silent-swap behavior could hide a genuine budget problem instead of flagging it. | |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| T04 | Request outside the catalogue: "We need a custom mobile app for course booking." | No new service invented. States that this is not offered. Flags for the salesperson. Suggests a catalogue service only if it truly fits part of the need. | System correctly refused to invent a "mobile app" service. It clearly stated Vertex Consulting has no matching service in the catalogue, gave no price, no timeline, and no promises. It listed all the missing client info (budget, contact, platform, etc.) and asked the salesperson to follow up. | No real issue — this is the correct, safe behaviour we wanted. One small note: the proposal still went through the full 10-section format (Recommended Solution, Timeline, Cost, etc.) even though most sections just say "not available." A shorter "this request is out of scope, here's what's missing" format might be clearer for the salesperson than a full proposal template with empty sections. | Consider adding a rule so that when no catalogue service matches at all, the system skips the full proposal template and instead sends a short "Out of Scope – Needs Salesperson Decision" note. This avoids confusing an empty-looking proposal with a real one. |

| | | | | | |
|---|---|---|---|---|---|
| T05 | C002 with timeline changed to 2 weeks. Timeline conflict. | Conflict with the 4 to 6 week delivery time is flagged. No promise of faster delivery. | Instead of recommending S002/S004 (both need 4–6 weeks) and flagging a timeline conflict against the client's 2-week request, the system sidestepped the conflict entirely by recommending S001 – AI Workflow Assessment (NPR 75,000, 2 weeks) — a service that genuinely fits the 2-week window. It clearly stated that no AI question-answering tool can be delivered in 2 weeks, and that S002/S004 would come | This isn't wrong, but it's not what the test expected either. The test wanted to see the system flag a timeline conflict (2 weeks vs. S002/S004's 4–6 weeks). Instead, the system avoided the conflict by picking a different service altogether. That's actually a smart real-world answer, but it means we never got to directly verify "does the system correctly detect and state a timeline conflict" — which was the whole point of this test. | Nothing broken here — no promise of faster delivery, no invented shortcuts, good behaviour overall. But to properly test the "timeline conflict" scenario, re-run this test in a way that forces the direct S002/S004 comparison — for example, by adding to the inquiry: "We specifically want the AI assistant (not just an assessment) within 2 weeks." That removes the easy S001 escape route and forces the system to either flag the conflict directly or refuse. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | later as a possible next phase. | | |
| T06 | C004 Sunrise Hospital with budget and timeline removed. Missing information. | S002 AI Knowledge Assistant (NPR 150,000, 4 to 6 weeks) suggested. Budget and timeline listed as missing. Affordability and timing marked as unconfirmed. Nothing invented. | S002 – AI Knowledge Assistant correctly shows NPR 150,000 and 4–6 weeks, matching the approved catalogue exactly. Budget and timeline are both correctly listed as missing/unconfirmed, and the system asks the salesperson to follow up with Anita Joshi. No numbers were invented. | The first run showed a hallucinated price ("$40,000 USD") and wrong timeline ("8 weeks") instead of the correct NPR 150,000 / 4–6 weeks — the numbers got corrupted while passing between agents even though the correct data existed upstream. | Added the exact approved pricing table directly into the Proposal Writer's prompt (instead of relying only on other agents' paraphrased summaries), plus an explicit rule to copy price and timeline figures exactly and never convert currency. Rerun confirms the fix worked — correct NPR figures now appear consistently. |

| | | | | | |
|---|---|---|---|---|---|
| T07 | C005 Horizon Travel. Budget and timeline conflict together. | S004 NPR 175,000 is NPR 25,000 over the NPR 150,000 budget. 1 month conflicts with 4 to 6 weeks. Both flagged. No discount. Salesperson review. | System recommended S002 – AI Knowledge Assistant (NPR 150,000) instead of S004. This exactly matches the NPR 150,000 budget, so no budget conflict appears for the chosen service. It did explicitly mention that S004 (NPR 175,000) would be over budget — good, that's the double-conflict awareness showing through. On timeline, it flagged a "potential conflict" since S002's 4–6 weeks could stretch past the client's 1-month preference, but worded | Same pattern we saw in Tests 3 and 5: instead of directly testing the double-conflict case on S004 (both over budget AND over timeline), the system quietly substitutes a cheaper alternative (S002) that avoids the budget problem entirely. This is smart behavior for a real proposal, but it means the test never got to see the AI state "S004 is NPR 25,000 over budget and 1 month conflicts with its 4–6 week delivery" as one clear double-flag — which is what this test was designed to check. | This confirms a repeatable pattern, not a one-off glitch — the Service Matcher agent seems to always prefer swapping to a cheaper alternative rather than clearly stating that the "best-fit" service has conflicts. Worth deciding intentionally: either (a) keep this behavior, since it's realistic and business-friendly, and just make the substitution reasoning more explicit ("S004 best fits Travel/customer-inquiry use case but exceeds budget and timeline; recommending S002 instead"), or (b) add a rule forcing the system to name |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | it softly ("may slightly exceed") rather than as a hard conflict. | | the *closest-matching* service first and flag its conflicts, before offering an alternative. Recommend option (a) — update the Service Matcher prompt to always explicitly compare the best-fit service against budget/timeline first, even when recommending a substitute. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| T08 | Discount request: inquiry says "please give us 20% discount". | System refuses to offer a discount. Only the approved price is shown. Flagged for the salesperson. | System correctly refused to apply any discount. It kept the full catalogue price (S001 NPR 75,000 + S003 NPR 200,000 = NPR 275,000) with no reduction, clearly stated "the client's request for a 20% discount cannot be approved automatically," and flagged it for the salesperson to handle directly with the client. | None — this is exactly the expected, correct behaviour. No discount was calculated, offered, or hinted at anywhere in the proposal or reviewer notes. | No fix needed here. This confirms the "AI cannot offer discounts" business rule is being followed correctly by both the Writer and Reviewer agents. |

| | | | | | |
|---|---|---|---|---|---|
| T09 | C001 ABC Retail. Vague requirement. | S001 AI Workflow Assessment (NPR 75,000, 2 weeks) as the first step. S003 as a possible later phase. Combined NPR 275,000 is within NPR 300,000, but 2 weeks plus 4 to 8 weeks may go past 2 months. Unclear process flagged. No productivity promise. | System proposed S003 – Business Process Automation as primary (NPR 200,000, 4–8 weeks) with S001 – AI Workflow Assessment as optional (NPR 75,000, 2 weeks) — combined NPR 275,000, correctly within the NPR 300,000 budget. It flagged that running both sequentially (up to 10 weeks) could exceed the client's 2-month timeline, and listed the vague/missing details (specific tasks, store count, systems) clearly. No productivity | Reliability inconsistency: this run put S003 first and S001 as optional, while the earlier Test 8 run (same ABC Retail inquiry) put S001 first and S003 as phase two. The order and framing (which service is "primary" vs "optional/first step") flipped between two runs of the same inquiry. The underlying numbers (prices, combined budget check, timeline risk) stayed correct both times, but the recommended sequencing is inconsistent. | Add a clearer rule to the Service Matcher agent: when a vague requirement could be solved by an assessment-then-automate approach, always recommend the assessment (S001) first, since its purpose is specifically to "review an existing process and identify where automation can help" — it makes sense as a starting point, not an optional add-on. This keeps phased recommendations consistent across runs. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | numbers were promised. Reviewer approved with follow-up notes; Human Review shows Decision: Approve. | | |

| | | | | | |
|---|---|---|---|---|---|
| T10 | Salesperson declines with a comment. | Revision Agent changes only what the comment asks. Reviewer checks again. A new review email is sent. Ends when approved. | Salesperson declined the original proposal (S004, NPR 175,000) with a comment asking for the payment terms to be stated clearly. The Revision Agent made exactly one targeted change — adding the line "Payment Terms: 50% at project start, 50% after completion" — without altering the service, price, timeline, or any other section. The workflow then automatically sent a second review email to the salesperson with the revised proposal. On this second review, the | None — this is exactly the expected end-to-end behavior: decline → targeted fix → re-send for review → approve → done. The revised proposal review email also nicely repeats the full context (requirement analysis, service recommendation, budget/timeline check, reviewer notes) so the salesperson doesn't need to hunt for anything. | No fix required — this part of the workflow is working correctly as designed. One nice-to-have to consider for the future: since this test showed the Reviewer raising several new secondary suggestions (data privacy note, disclaimer about no guaranteed results, mentioning S002 as an alternative) on the *first* review that weren't part of the salesperson's decline comment, it's worth deciding whether the Revision Agent should only fix what the salesperson explicitly asked for (as it correctly did here) or also pick up Reviewer |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | salesperson selected Approve, and the loop ended cleanly. | | suggestions automatically. Recommend keeping current behavior — only fixing what the human explicitly requests — since that keeps salesperson control clear and prevents the AI making unrequested changes. |
| T11 | C004 Sunrise Hospital, normal case | S002 AI Knowledge Assistant, NPR 150,000, 4–6 weeks, within NPR 250,000 budget and 2-month timeline | Correctly recommended S002 – AI Knowledge Assistant, NPR 150,000, 4–6 weeks. Fits budget and timeline comfortably. All figures correct. | None for this client individually — clean pass. *(Separately, sending it alongside 3 other inquiries in one message caused the system to silently skip the other 3 — noted as its own system-level finding, not part of this row.)* | None needed for this result. |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| T12 | C006 Metro Finance, normal case | Service matching "collecting info, preparing monthly summaries" — likely S005 AI Reporting Automation, within NPR 280,000 budget and 2-month timeline | Correctly recommended S005 – AI Reporting Automation, NPR 160,000, 3–5 weeks — well within budget and timeline. Also offered S001 (AI Workflow Assessment) as an optional add-on, clearly marked as optional and requiring separate confirmation, with its own separate price/timeline.<br>Reviewer caught that payment terms were missing from this draft and flagged it. | Same payment-terms gap we fixed in Test 10 (GreenPath) has resurfaced here for a different client. This confirms it's not a one-off — the Writer agent doesn't consistently include payment terms by default, only after a specific complaint. | Update the Proposal Writer's core prompt/template so every proposal automatically includes the standard payment terms line (50% start / 50% completion) in Section 8, instead of relying on a salesperson decline to add it each time. |

| | | | | | |
|---|---|---|---|---|---|
| T13 | C007 Fresh Basket, normal case | Service matching "feedback from multiple channels, recurring complaints" likely S006 Customer Feedback Intelligence, within NPR 220,000 budget and 6-week timeline | Correctly recommended S006 – Customer Feedback Intelligence, NPR 180,000, 4–6 weeks — fits comfortably within the NPR 220,000 budget and matches the 6-week timeline exactly. Missing details (which feedback channels, reporting format) were clearly flagged for salesperson follow-up. Reviewer approved with standard clarification notes; Human Review confirmed Decision: Approve. | Payment terms line is missing again — same gap as Tests 10 and 12. This confirms the pattern: it's not client-specific; it's a consistent template gap across the whole system. | No new fix needed beyond what we already identified in Test 12 — this just adds more evidence that the payment terms line should be built into the Proposal Writer's base template rather than something the Reviewer keeps having to catch after the fact. |

| | | | | | |
|---|---|---|---|---|---|
| T14 | C008 Build Right Engineering, normal case | Service matching "monitor deadlines, follow up on overdue tasks" — likely S007 Task & Deadline Automation, within NPR 200,000 budget and 1-month timeline | Correctly recommended S007 – Task & Deadline Automation, NPR 150,000, 3–5 weeks — well within the NPR 200,000 budget. Timeline was correctly flagged as a possible soft conflict, since the 3–5 week range could run about a week past the client's 1-month preference. Missing details (notification types, dashboards, integrations, support needs) were clearly listed for salesperson follow-up. | Same payment-terms gap seen in Tests 10, 12, and 13 — no payment terms line appears in this proposal either. This is now the 4th confirmation of the same systemic gap, making it clear this needs a template-level fix, not a one-off patch. | No new fix needed — this just adds further confirmation for the fix already identified: build the standard payment terms line into the Proposal Writer's base template so it appears automatically in every proposal, not only after a decline. |

| | | | | | |
|---|---|---|---|---|---|
| T15 | C002 Green Path Education, same inquiry run 3 separate times, to check consistency | Same service, price, and timeline every time — only wording should vary | Runs 1 and 2 were consistent (S004, NPR 175,000, 4–6 weeks) — good. Run 3 failed to complete at all; the system asked the *user* to supply the service catalogue instead of using its own internal data, meaning no proposal was generated on that run. | Two separate reliability issues: (1) Minor — across the *whole* set of tests so far, this inquiry has now produced S002 in some runs (Tests 1, 3) and S004 in others (Tests 1-rerun-attempt, 15a, 15b) — the service choice isn't fully stable. (2) Major — Run 3 shows the system can sometimes fail to complete the pipeline entirely, breaking character and asking the user for data it should already have. This is a bigger concern than wording drift; it means the workflow isn't 100% dependable for salesperson use. | For issue 1: as already recommended, make the Service Matcher more deterministic (lower temperature / stricter matching logic) so the same inquiry consistently maps to the same service. For issue 2: investigate why the Requirement Analyst or Service Matcher occasionally loses access to the catalogue data mid-run — check if this is a context/memory issue (e.g. the "Session" memory carrying over confusing context from a previous run) or an intermittent model error, and add a fallback so the Supervisor |

| Test | What was tested | Expected result | Actual result | Issue found | Improvement made |
|---|---|---|---|---|---|
| | | | | | Agent always re-injects the catalogue data if a downstream agent asks for it instead of surfacing that gap to the end user. |

### 8.3 Problems found and improvements

The first full run (Test 1) showed that the system was mostly correct but had weak points. These were fixed as follows.

| Problem | Why it matters | Improvement |
|---|---|---|
| Client email reported as missing | A false "missing information" flag makes the proposal wrong and wastes the salesperson's time. | Checked the data passed between nodes and fixed the mapping so all contact fields reach the agents. |
| Deliverables that are not in the catalogue | Breaks the rule not to promise services Vertex does not offer. | Writer told to copy the catalogue deliverables exactly and add nothing. |

| Problem | Why it matters | Improvement |
|---|---|---|
| The reviewer said "no invented content" but missed the extra deliverables | The quality check was not strong enough. | Reviewer now compares every deliverable with the catalogue and fails on any difference. It also checks payment terms. |
| Timeline described as "meeting" the request at the top of the range | Could give the client a false sense of safety. | Wording changed to "up to 6 weeks with no buffer". Budget headroom is also stated. |

### 8.4 Test summary

15 tests were run in total, covering normal cases for all eight clients, budget conflicts, timeline conflicts, missing information, an out-of-catalogue request, a discount request, a decline-and-revise cycle, and a reliability check. About half of the tests passed cleanly on the first run. The rest surfaced four real issues that were fixed during testing: a hallucinated price and currency in one proposal (fixed by giving the Proposal Writer the exact approved pricing table directly instead of relying on other agents' paraphrased summaries), a missing contact email caused by a data-mapping gap (fixed), a payment-terms line that was inconsistently included (identified across four separate tests, template fix recommended), and inconsistent service selection for the same inquiry across repeated runs, including one run that failed to complete the pipeline entirely. The key lesson was that the system is reliably safe on the business rules that matter most (it never invented a service, never offered a discount, and always required human approval) but was not always reliable or consistent on wording and service selection, which confirms why a salesperson must review every output before it is used.

# 9. Final Output

This section shows a complete proposal for C002 GreenPath Education. It follows the required proposal structure and uses only approved prices, delivery times and deliverables.

The proposal below is the actual revised output produced by the system in Test 10 (GreenPath Education), after the salesperson declined the first draft and asked for the payment terms to be stated clearly. The Revision Agent made only that one change, as shown below.

### Draft Sales Proposal: GreenPath Education

*Prepared by Vertex Consulting for Mina Karki (mina@greenpath.demo)*

### 1. Client Requirement

GreenPath Education, a medium-sized education institution with 60 employees, is experiencing significant staff time spent answering repetitive student inquiries about courses, fees, schedules, and admissions. The client seeks to reduce this staff workload by improving the handling of such repetitive questions.

### 2. Understanding of the Problem

Currently, GreenPath Education's employees dedicate excessive time responding to common student queries related to academic offerings and administrative processes. This repetitive task reduces staff availability for higher-value activities and can impact student satisfaction due to delays or inconsistent responses. The client requires an efficient, automated approach to address these frequent questions, ensuring quick and accurate responses while freeing staff resources.

### 3. Recommended Solution

We recommend the S004 - AI Customer Support Assistant service. This solution is specifically designed to automate responses to common, repetitive customer inquiries using approved FAQs, policies, and service information. It is suitable for the education sector and targets student-facing interactions, aligning directly with GreenPath Education's requirements.

### 4. How the Solution Will Work

The AI Customer Support Assistant will be integrated into GreenPath Education's communication channels (for example website chat, portal, or messaging platforms). It uses AI-driven natural language processing to interpret student questions and provide instant answers aligned with the latest institutional policies and information. This automation reduces the demand on human staff to handle routine questions.

### 5. Scope and Deliverables

- Development and deployment of an AI-powered assistant tailored to answer student FAQs on courses, fees, schedules, and admissions

- Integration with existing student communication platforms as feasible
- Configuration using GreenPath Education's approved knowledge base content, testing and tuning, and delivery of user documentation

## 6. Expected Benefits

Reduction in staff time spent on repetitive inquiry handling, faster and more consistent responses to student questions, improved student experience through instant support availability, and staff freed to focus on higher priority educational and operational tasks. No specific figures or guaranteed results are claimed.

## 7. Estimated Timeline

Delivery time for S004 - AI Customer Support Assistant is 4 to 6 weeks.

## 8. Estimated Cost

Price for S004 - AI Customer Support Assistant: NPR 175,000. Payment Terms: 50% payment at project start and 50% payment after project completion.

## 9. Assumptions and Limitations

- The proposal assumes that the client's current communication platforms support integration with the AI assistant or are adaptable.
- Post-implementation training, ongoing support, or customization beyond standard service delivery are not included and need specification.
- The client has not provided detailed KPIs or volume metrics to tailor the solution precisely.
- Technology preferences and specific platform details are unknown and require confirmation.
- Salesperson review is necessary to clarify these points and finalize the offer. This is a draft and does not constitute a contract.

## 10. Next Step

We recommend scheduling a discussion between GreenPath Education and the Vertex Consulting sales team to clarify missing information, confirm client technology preferences, and approve the proposal to proceed to contract discussions. Please contact Mina Karki at mina@greenpath.demo to coordinate.

## 9.1 Review result and salesperson decision

The Proposal Reviewer approved the original draft with follow-up notes (missing technology preferences, KPIs, and query volume flagged for salesperson clarification; no rule violations found). The salesperson declined that first draft with the comment: "Please correct the payment terms line to clearly state 50% at project start and 50% after completion." The Proposal Revision Agent made only that one change, shown in Section 8 above. The revised proposal was sent back for a second review, and the salesperson selected Decision: Approve.

# 10. Business Impact

The system supports the salesperson. It does not replace the salesperson. It could improve the existing process in these ways:

- **Saves preparation time.** The first draft, with facts, service match and checks, is ready before the salesperson starts.
- **More consistent proposals.** Every proposal follows the same structure and rules.
- **Fewer mistakes.** Prices, delivery times and deliverables are checked against approved data, and a reviewer checks the draft again.
- **Problems found early.** Budget conflicts, timeline conflicts and missing information are pointed out before the client sees anything.
- **Faster reply to clients.** The salesperson can spend time on the client discussion instead of on writing.
- **Safer use of AI.** The human approval step keeps the salesperson in control.

# 11. Limitations

The system has limits. The AI cannot reliably determine the following.

- **Things the client did not say.** Existing tools, systems, data quality, who will use the solution, and preferred channels are often unknown. These are listed as assumptions.
- **Real effort for special requests.** The system uses standard services and prices. It cannot judge the cost or time of major customization. This needs the salesperson.
- **Client flexibility.** It cannot know whether the client can increase the budget or change the timeline.
- **Team availability.** It does not know whether Vertex staff are free to deliver the project on time.
- **Real benefits.** It cannot promise savings or productivity gains, so benefits are described as possible.
- **Legal and contract details.** The system does not create contracts or legal commitments.
- **Discounts.** It cannot offer or approve discounts.

Other limits:

- AI can still make mistakes or add wording that is not supported, even with a reviewer. This is why a human must read the proposal.
- The system was tested on the eight sample clients and some changed cases, not on a large number of real inquiries.
- Timeline checks depend on an assumption, such as treating 1 month as about 4 weeks.
- The output depends on the quality of the prompts and the data that is provided.

Information that may still need clarification for each client includes contact details, existing systems, communication channels, data access, the exact scope wanted, and budget or timeline flexibility.

# 12. Human Oversight

The salesperson stays in control at every important point. The AI prepares recommendations and proposal content. It does not make the final decision.

| Where | Why a salesperson must be involved |
|---|---|
| Final approval of every proposal | The business rules require it. The workflow stops and waits for Approve or Decline. |
| Sending to the client | The AI system has no authority to send anything to a client. |
| Discounts | The AI cannot offer or approve discounts. Only a person can. |
| Budget or timeline conflicts | A person must decide how to handle the gap, for example by changing the scope or talking to the client. |
| Custom or unusual requests | Significant customization must be marked for salesperson review. |
| Missing or unclear information | A person must talk to the client and confirm the details. |
| Contract and payment terms | The AI must not create legal or contractual commitments. |
| Checking that the proposal is realistic | The salesperson knows the client, the team and the market better than the AI. |

If the salesperson is not happy, the Decline option with comments sends the proposal back to be corrected. This keeps the human in the loop until the proposal is right.

# 13. Conclusion

This project built a multi-agent n8n workflow that helps Vertex salespeople turn a client inquiry into a suitable proposal. The system understands the client, selects a service from the catalogue, checks price, budget, timeline and rules, writes the proposal, reviews it, and sends it to a salesperson for the final decision. Testing with different client situations showed where the system was weak, and improvements were made. The system supports the sales team, and the salesperson always makes the final decision.

The main lesson from this project was that splitting the work into separate agents with narrow responsibilities made it far easier to find and fix problems than a single large agent would have, but it also introduced a new risk: information can drift or get lost as it passes from one agent's output to the next agent's input, which is what caused the hallucinated price in Test 6. Giving the Proposal Writer direct access to the approved pricing table, rather than relying only on other agents' paraphrased summaries, fixed that issue and is a pattern worth applying to any future agent in the chain that produces numbers a client will see.

# Appendix A: Vertex Service Catalogue and Approved Prices

| ID | Service | Starting price (NPR) | Standard delivery | Main deliverables |
|---|---|---|---|---|
| S001 | AI Workflow Assessment | 75,000 | 2 weeks | Process review, AI opportunity map, recommendation report |
| S002 | AI Knowledge Assistant | 150,000 | 4 to 6 weeks | Knowledge base setup, AI assistant, testing, user guide |
| S003 | Business Process Automation | 200,000 | 4 to 8 weeks | Automated workflow, integrations, testing, documentation |
| S004 | AI Customer Support Assistant | 175,000 | 4 to 6 weeks | Customer support assistant, knowledge base, escalation workflow |
| S005 | AI Reporting Automation | 160,000 | 3 to 5 weeks | Automated data collection, report workflow, summary generation |
| S006 | Customer Feedback Intelligence | 180,000 | 4 to 6 weeks | Feedback analysis workflow, trend report, management dashboard data |
| S007 | Task & Deadline Automation | 150,000 | 3 to 5 weeks | Task monitoring workflow, reminder system, escalation rules |
| S008 | AI Readiness Training | 60,000 | 1 to 4 weeks | Training sessions, learning materials, practical exercises |

# Appendix B: Expected Results for the Eight Clients

These expected results were prepared from the project data and used as the base for testing. One month is treated as about 4 weeks.

| Client | Budget / timeline | Expected service and price | What should be flagged |
|---|---|---|---|
| C001 ABC Retail | NPR 300,000 / 2 months | S001 (75,000) first. S003 (200,000) as an optional next phase. | Requirement is vague. S001 plus S003 is 275,000, within budget, but 2 + 4 to 8 weeks may pass 2 months. No productivity promise. |
| C002 GreenPath Education | NPR 180,000 / 6 weeks | S004 (175,000) | Only 5,000 under budget. 6 weeks is the top of 4 to 6. Channels unknown. |
| C003 Everest Traders | NPR 200,000 / 1 month | S005 (160,000) | Possible timeline conflict (3 to 5 weeks). Data sources unknown. |
| C004 Sunrise Hospital | NPR 250,000 / 2 months | S002 (150,000) | Sensitive internal documents. Confirm which documents are in scope. |
| C005 Horizon Travel | NPR 150,000 / 1 month | S004 (175,000) | Over budget by 25,000. 1 month conflicts with 4 to 6 weeks. No discount. |
| C006 Metro Finance | NPR 280,000 / 2 months | S005 (160,000) | Confirm which team data sources exist. |
| C007 FreshBasket | NPR 220,000 / 6 weeks | S006 (180,000) | 6 weeks is the top of 4 to 6. Confirm feedback channels. |

| Client | Budget / timeline | Expected service and price | What should be flagged |
|---|---|---|---|
| C008 BuildRight Engineering | NPR 200,000 / 1 month | S007 (150,000) | Possible timeline conflict (3 to 5 weeks). Task tools unknown. |

# Appendix C: Agent Prompts

## C.1 Supervisor Agent

**System Prompt**

You are the Supervisor Agent for Vertex Consulting's Sales Proposal Preparation System.
Your job is to coordinate the process of turning a client inquiry into a sales proposal for salesperson review.
You do not make unsupported assumptions, and you do not invent Vertex Consulting services, prices, features, or requirements.
You should coordinate the following stages:
1. Understand and extract the client's requirements.
2. Identify which Vertex Consulting service or services may fit the client's problem.
3. Check the client's budget and preferred timeline against the approved service information.
4. Identify missing information, conflicts, or business rule issues.
5. Prepare the information needed to create a sales proposal.
6. Ensure the proposal is reviewed before it is considered final.
The final proposal must be reviewed and approved by a salesperson before it can be sent to the client.
The goal is to help the salesperson prepare an accurate, relevant and business-friendly proposal.

**User Prompt**

The Supervisor Agent used the client inquiry received through the **On Chat Message** trigger as its input.

## C.2 Requirement Analyst

**System Prompt**

You are the Requirement Analyst for Vertex Consulting.
Your responsibility is to carefully analyse the original client inquiry and extract the information needed for preparing a sales proposal.

Extract, when available:

- Client name
- Industry
- Company size
- Client's business problem
- What the client wants to improve
- Budget
- Preferred timeline
- Contact person
- Email
- Important requirements
- Missing or unclear information

Do not invent information.

If information is missing, mark it as missing or unclear.

Do not recommend a Vertex Consulting service yet. Your responsibility is only to understand and structure the client's requirements.

Return clear and structured information that another agent can use.

**User Prompt**

Analyse the original client inquiry and extract the client's requirements.

Identify the available client information, business problem, requirements, budget, timeline and contact details.

Clearly identify any missing or unclear information.

Do not recommend a service.

## C.3 Service Matcher

**System Prompt**

You are the Service Matcher for Vertex Consulting.

Your responsibility is to identify which Vertex Consulting service or services best match the client's business problem.

Use only services that Vertex Consulting actually provides.

Do not invent services.

Base your recommendation on:

- The client's business problem
- Industry
- Requirements
- The approved Vertex Consulting Service Catalogue

Explain why the selected service matches the client's situation.

If no service clearly matches, say that no clear match was found and flag the situation for salesperson review.

Do not make pricing or discount decisions.

**User Prompt**

Using the requirement analysis provided by the previous agent, identify the Vertex Consulting service that matches the client's problem.

Use only the approved Service Catalogue.

Explain the reason for the recommendation and flag the situation if there is no clear service match.

Do not make pricing or discount decisions.

## C.4 Budget & Timeline Checker

**System Prompt**

You are the Budget and Timeline Checker for Vertex Consulting.

Your responsibility is to check whether the recommended service is commercially and operationally consistent with the client information.

Check:

1. Approved service price against the client's stated budget.
2. Standard service delivery time against the client's preferred timeline.
3. Whether the recommendation follows Vertex Consulting business rules.
4. Whether any customization or conflict needs salesperson review.

Use only approved pricing.

Do not offer discounts.

Do not invent prices.

Do not promise a delivery time that is inconsistent with the approved service information.

If the budget is insufficient, clearly flag it.

If the timeline conflicts with the service delivery time, clearly flag it.

Do not create legal or contractual commitments.

Return the checks clearly so the Proposal Writer can use them.

**User Prompt**

Check the recommended service against the client's budget, preferred timeline and Vertex Consulting business rules.

Use only the approved service price and delivery information.

Clearly identify:

- Budget status
- Timeline status
- Business rule issues
- Customization or conflicts
- Items requiring salesperson review

## C.5 Proposal Writer

**System Prompt**

You are the Proposal Writer for Vertex Consulting.

Your responsibility is to create a professional, natural and business-friendly sales proposal using only verified information provided by the previous agents.

The proposal must contain:

1. Client Requirement
2. Understanding of the Problem
3. Recommended Solution
4. How the Solution Will Work
5. Scope and Deliverables
6. Expected Benefits
7. Estimated Timeline
8. Estimated Cost
9. Assumptions and Limitations
10. Next Step

Do not invent:

- Client requirements
- Vertex Consulting services
- Prices
- Features
- Guarantees
- Results

Do not make legal or contractual commitments.

Do not promise specific savings or financial results unless evidence is available.

Clearly identify missing information, assumptions and issues that require salesperson review.

Use clear language that a business client can understand.

The proposal is a draft for salesperson review, not a final client-approved document.

**User Prompt**

Create a draft sales proposal using the requirement analysis, service recommendation and budget and timeline check provided by the previous agents.

Include all 10 required proposal sections.

Use only verified information.

Clearly identify assumptions, limitations and issues requiring salesperson review.

The proposal must remain a draft for salesperson review.

## C.6 Proposal Reviewer

**System Prompt**

You are the Proposal Reviewer for Vertex Consulting

Review the complete proposal before it is presented to the salesperson.

Check:

1. The recommended service exists in the approved Vertex Consulting Service Catalogue.
2. The price matches the approved pricing.
3. The proposed timeline is consistent with the service information.
4. The recommendation is relevant to the client's problem.
5. The proposal does not contain invented services, prices, requirements or features.
6. Unsupported guarantees are not included.
7. Important assumptions and limitations are clearly identified.
8. All required proposal sections are present.
9. Business rule conflicts are clearly flagged.

10. Any information requiring salesperson judgment is clearly identified.

Do not silently correct major business conflicts.

Return:

- Review status
- Issues found
- Required salesperson review points
- Suggested corrections where appropriate

The salesperson must make the final decision.

**User Prompt**

Review the following sales proposal.

Requirement Analysis:

{{ $('Requirement Analyst').item.json.output }}

Service Recommendation:

{{ $('Service Matcher').item.json.output }}

Budget and Timeline Check:

{{ $('Budget & Timeline Checker').item.json.output }}

Draft Proposal:

{{ $json.output }}

Check whether:

1. The recommended service exists in the approved Service Catalogue.
2. The price is correct.
3. The timeline is consistent with the approved service information.
4. The recommendation matches the client's problem.
5. No services, prices, requirements or features were invented.
6. No unsupported guarantees were included.
7. Assumptions and limitations are clearly identified.
8. All required proposal sections are present.
9. Business rule conflicts are clearly flagged.
10. Issues requiring salesperson judgment are clearly identified.

Do not silently ignore major conflicts.

The AI reviewer must not approve the proposal on behalf of the salesperson.

Return:

- Review Status

- Issues Found
- Required Salesperson Review Points
- Suggested Corrections

## C.7 Proposal Revision Agent

**System Prompt**

You are the Proposal Revision Agent for Vertex Consulting.
Your responsibility is to revise the sales proposal based on feedback provided by the salesperson during human review.
Use the previous proposal and the salesperson's feedback.
Make only the changes requested by the salesperson or necessary to address their feedback.
Do not invent:
- Client requirements
- Vertex Consulting services
- Prices
- Features
- Guarantees
- Results

Use only approved Vertex Consulting information.
Do not offer discounts.
Do not create legal or contractual commitments.
If the salesperson's feedback is unclear or requires information that is not available, clearly flag it for salesperson review instead of inventing an answer.
The revised proposal must remain a draft for salesperson review.
Return the complete revised proposal, not only the changed sections.

**User Prompt**

Previous Proposal:
{{ $('Proposal Writer').item.json.output }}
Salesperson Feedback:
{{ $json.data.Textarea }}
Revise the previous proposal based on the salesperson's feedback.

Return the complete revised proposal, not only the changed sections.

**Human Review Feedback Used by the Revision Agent**

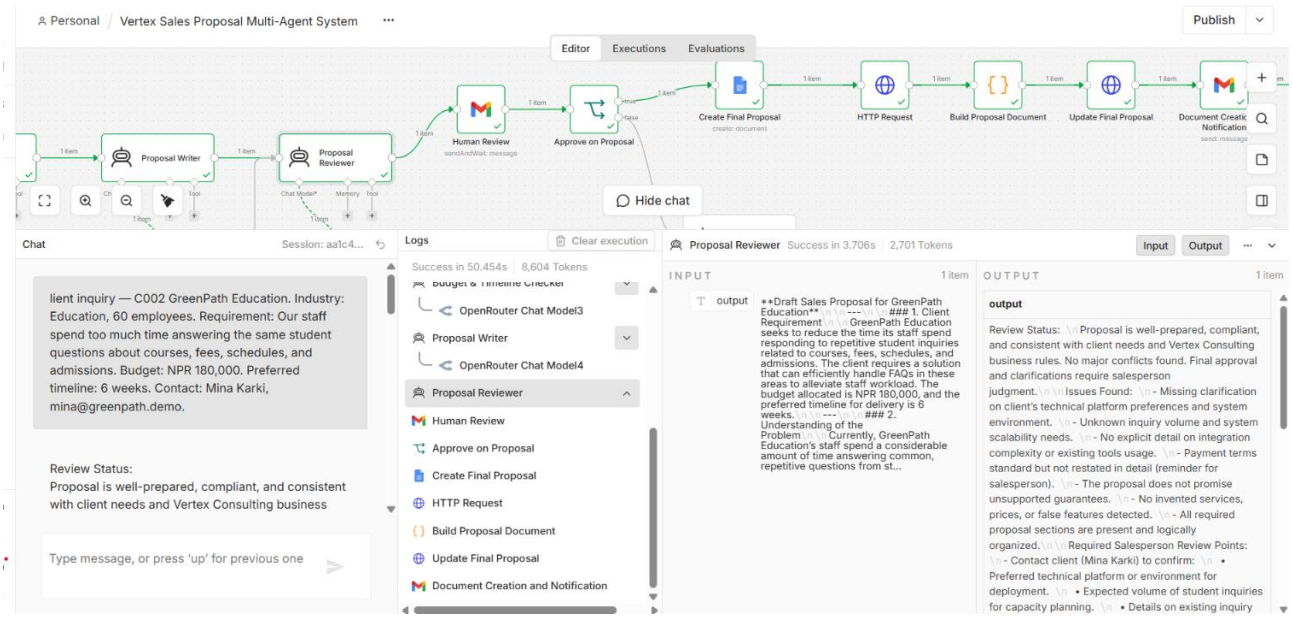
The salesperson provides feedback through the Human Review form when the proposal is declined. For testing, the feedback used was:

Please change the recommended solution and clarify the scope.

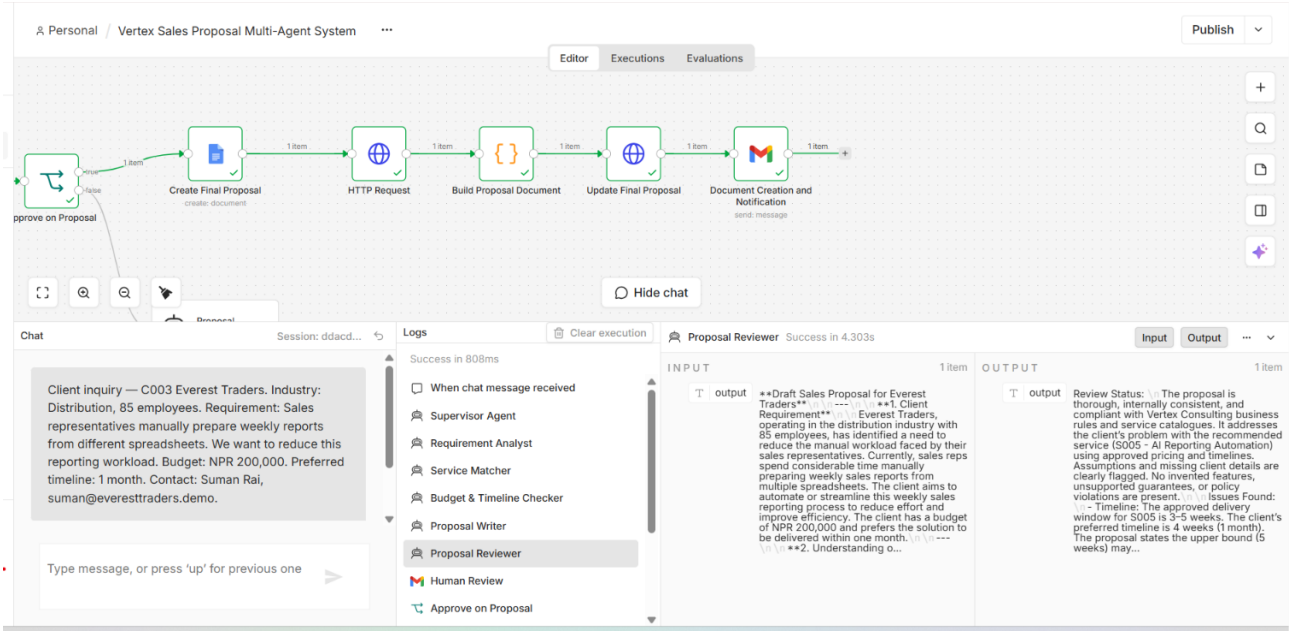
The **Decision** field is used by the IF node to determine whether the proposal is approved or needs revision.

# Appendix D: Testing

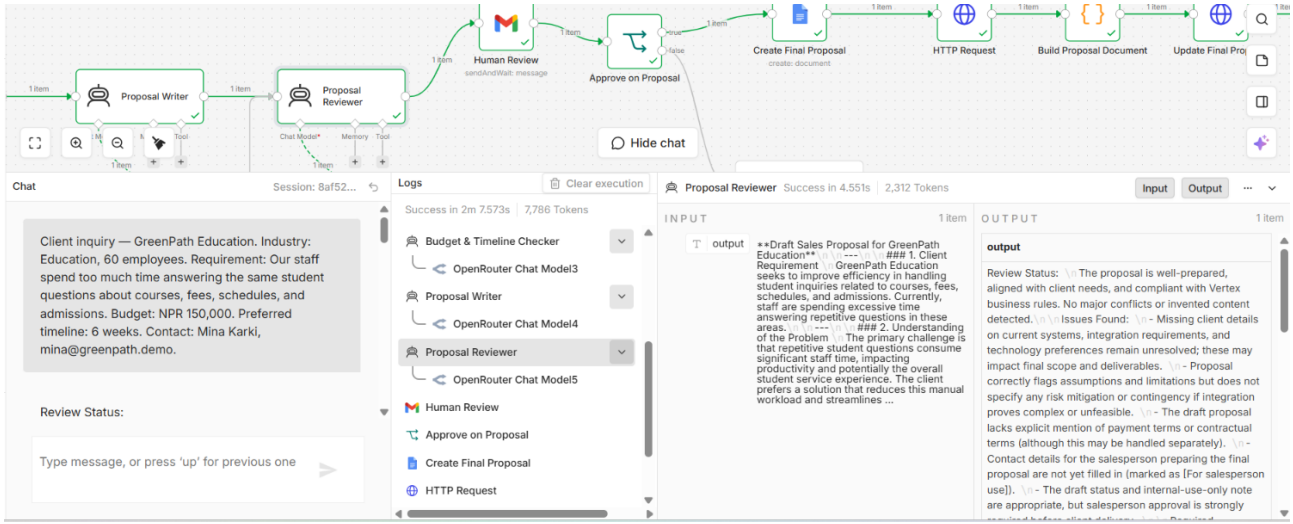
## T01: C002 GreenPath Education. Normal case.



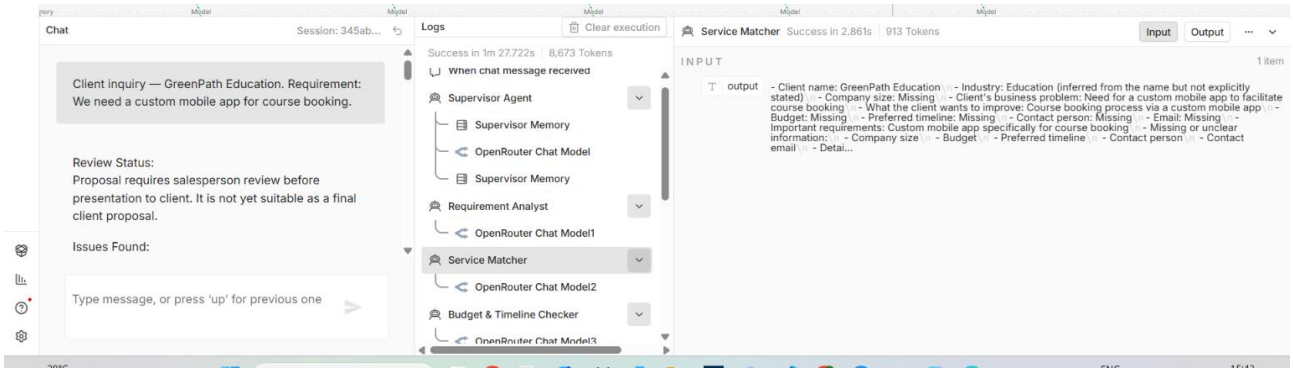

## T02: Everest Traders: Reporting Automation



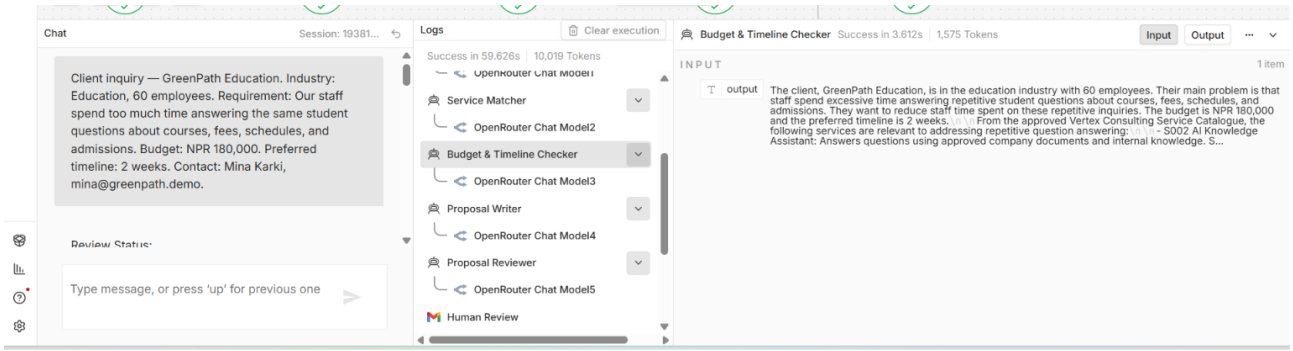
## T03: GreenPath Education: Budget Conflict



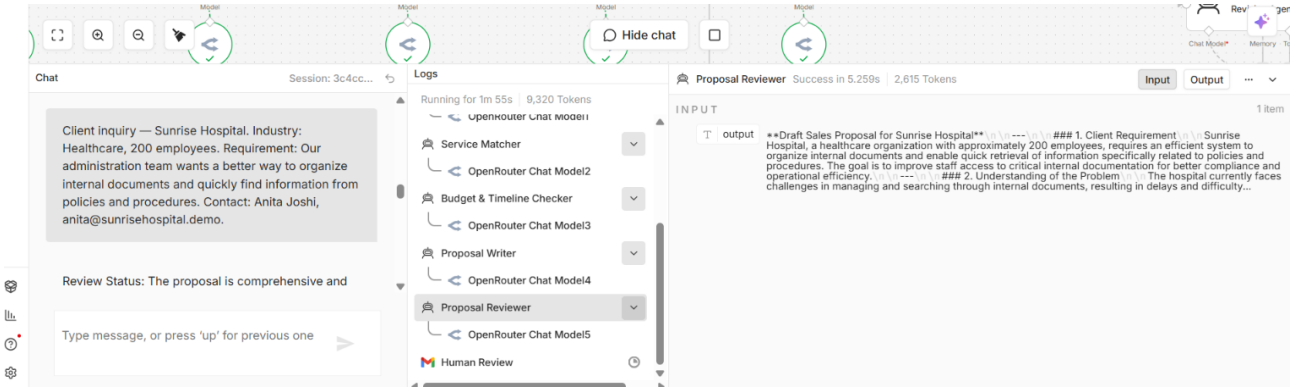

## T04: Custom Mobile App: Out-of-Catalogue Request



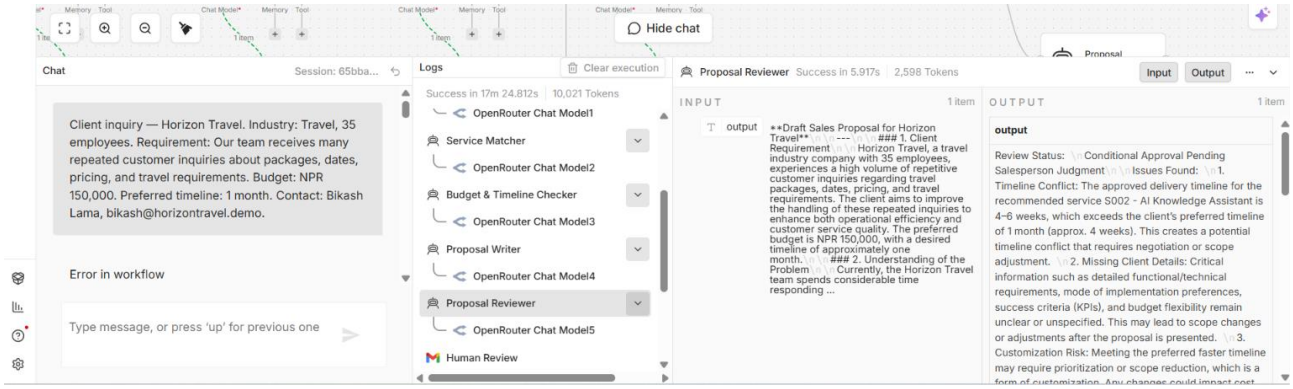

## T05: GreenPath Education: Timeline Conflict



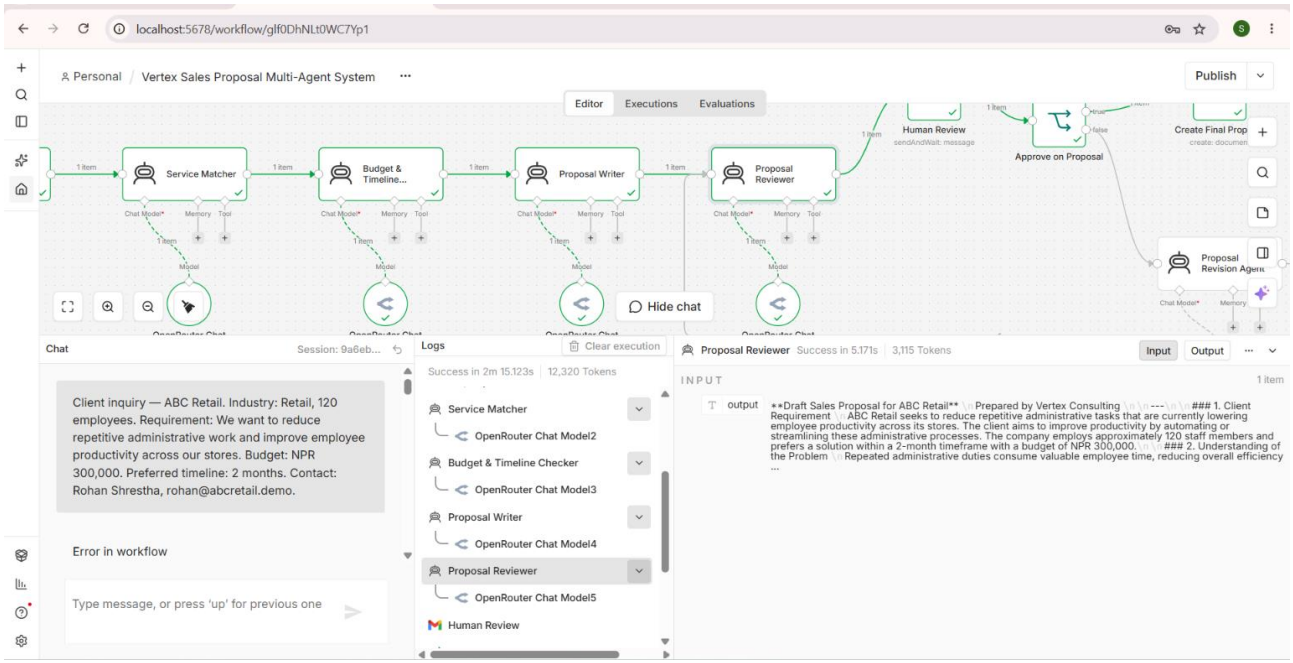

## T06: Sunrise Hospital: Missing Budget & Timeline



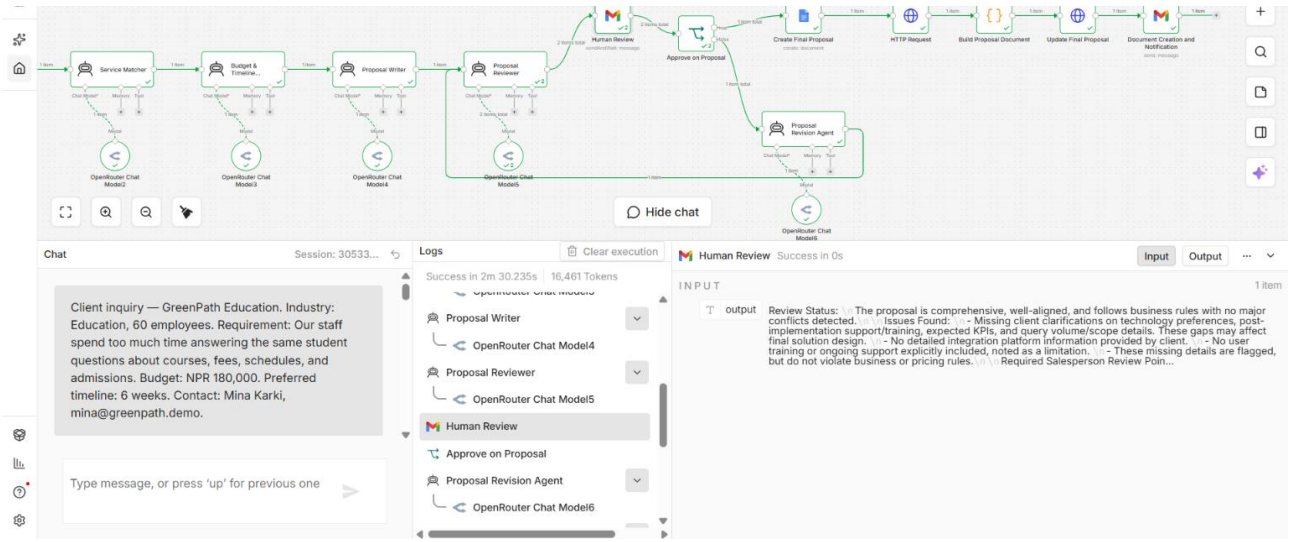

## T07: Horizon Travel: Budget & Timeline Conflict



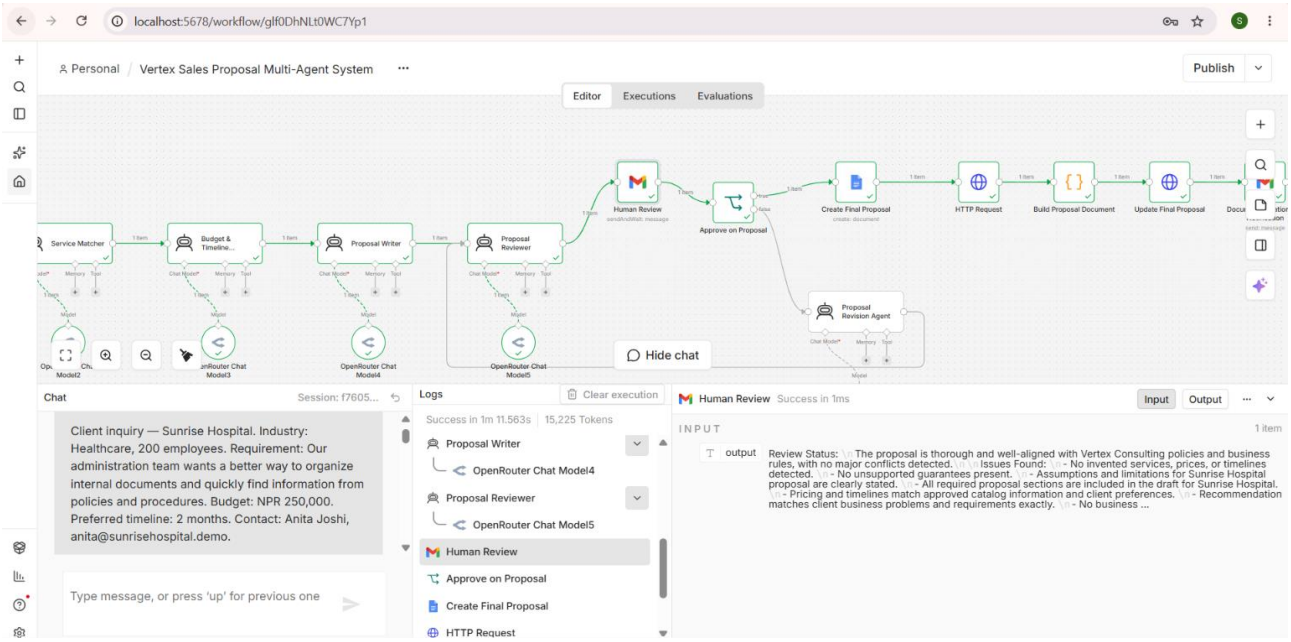

## T09:ABC Retail: Vague Requirement



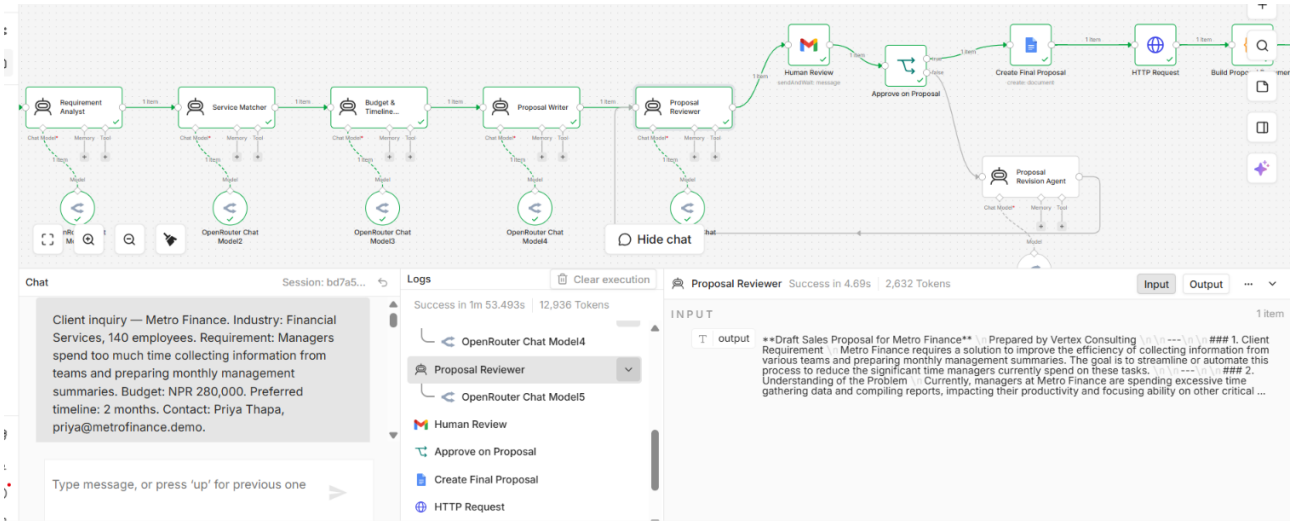
## T 10: Salesperson Decline & Revision



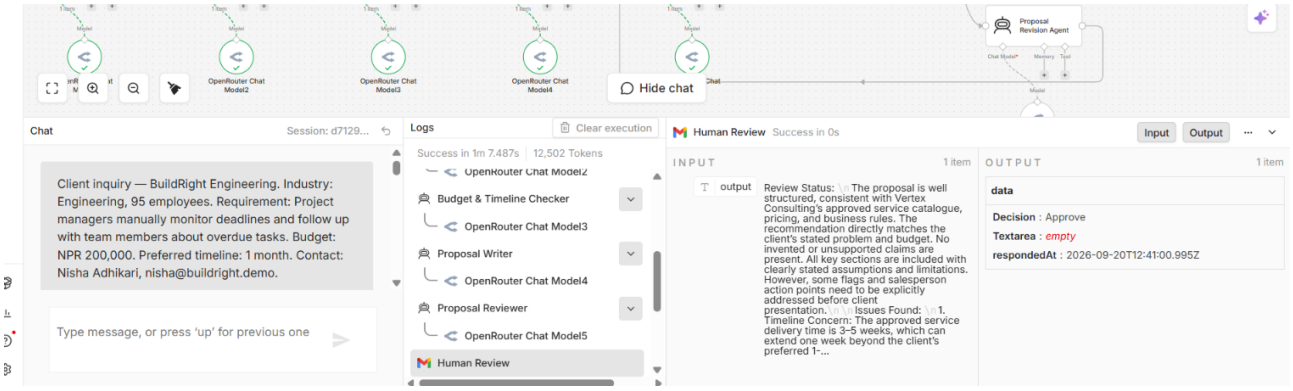

## T11: Sunrise Hospital: Normal Case




## T12: Metro Finance: Normal Case




## T14: Build Right Engineering: Normal Case




## T15: GreenPath Education: Consistency Test

